Physiology of Respiratory System



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Respiratory System

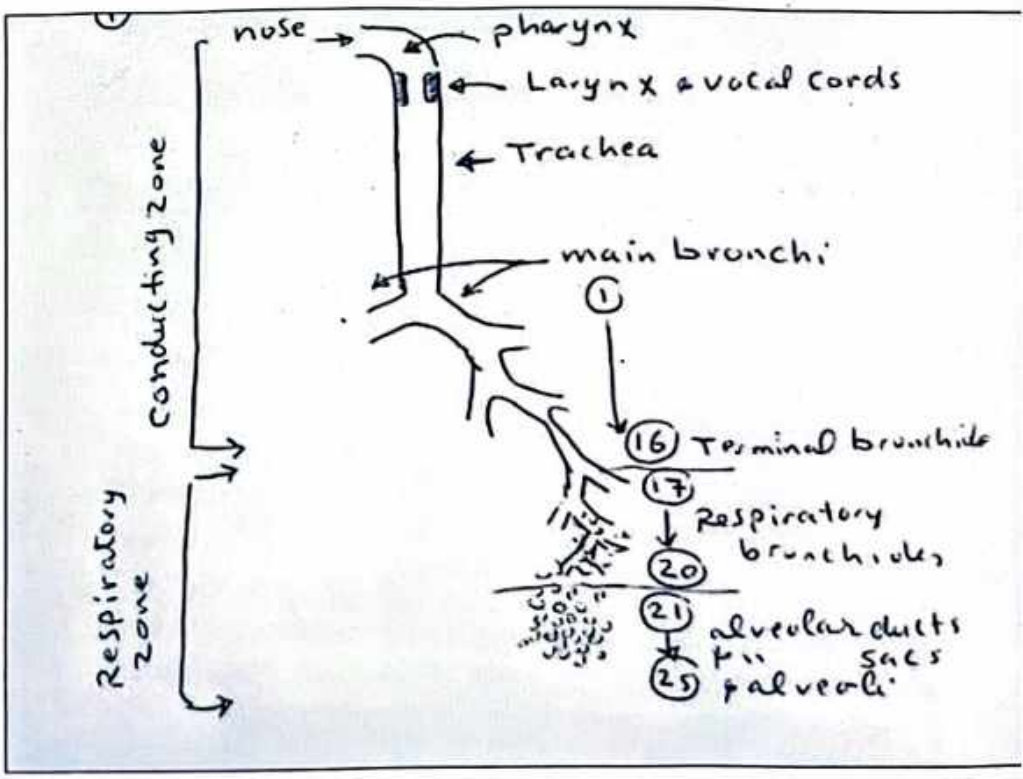
Upper respiratory tract: the nose, pharynx and larynx.

lower respiratory tract: the tracheobronchial tree,

which is divided into:.

Conducting zone:

- From trachea to terminal bronchioles.
- Does not contain alveoli.
- It conducts air at low resistance.
- It defends against microbes.
- It warms and moisturizes the air.
- Flow of air in the larynx causing vibration of the vocal cords producing sounds.

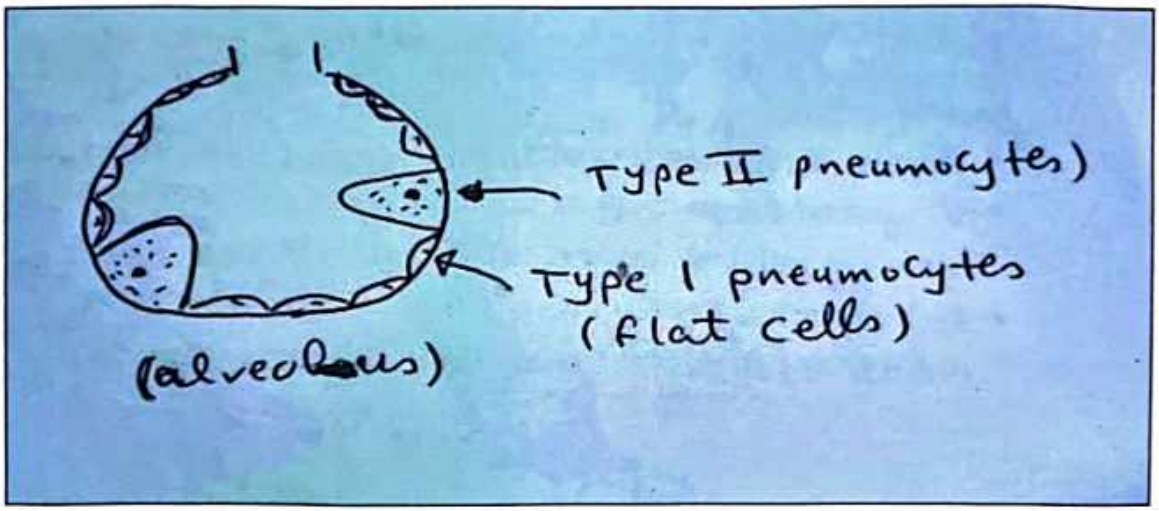


Respiratory zone:

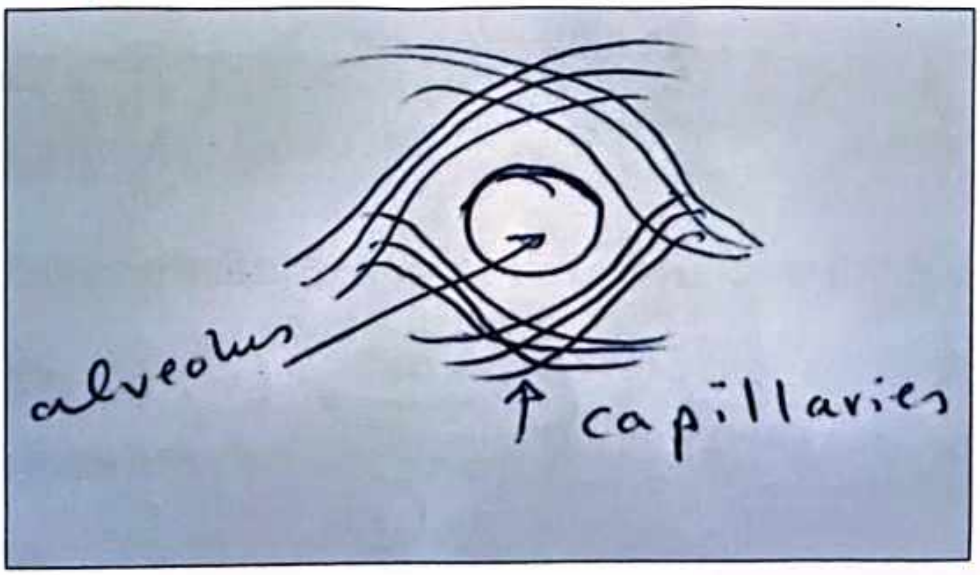
- Respiratory bronchioles, alveolar sac and alveoli.
- It contains alveoli (the site for gas exchange between air and blood).

The alveoli

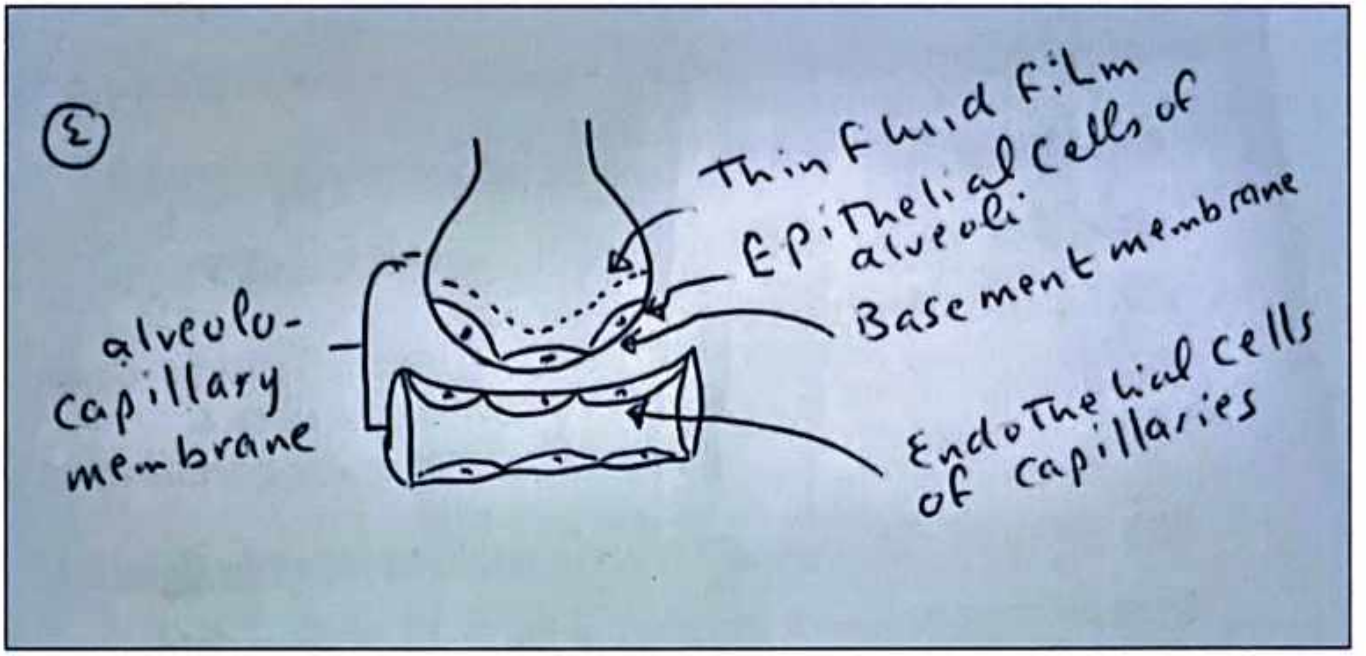
- About 300 million alveoli in both lungs (clusters of grape like sacs).
- Consist of single layer of flattened epithelial cells (TYPE I alveolar cells) → important for gas exchange.
- Also contains TYPE II alveolar cells which secret the pulmonary surfactant that facilitates lung expansion.



- Each alveolus is encircled by capillaries (1000 capillary/ alveolus) like a basket surrounding a ball.



The alveolar - capillary membrane:



- Thickness→ 0.5 um (very thin)

-The total surface area of alveoli → 70 m2.

- A double-walled closed sac called the pleural sac separates each lung from the thoracic wall and other surrounding structures .
- The interior of the pleural sac is known as the pleural cavity.
- The surfaces of the pleura secrete a thin intra-pleural, fluid which lubricates the pleural surfaces as they slide against each other during respiratory movement.

Functions of the respiratory system

A- Respiratory functions:

- Exchange of O_2 & CO_2 between the lungs and the blood.

B- Non respiratory functions:

1- Maintains acid base balance by altering the amount of CO_2 expired.

2- Enables speech (Vocalization).

3- Defends against inhaled foreign particles by presence of hair cells in the nose, cilia , Ig A, and mucus secretion in the tracheobronchial tree and phagocytes (PAM cells) at alvoli (for particles < 2 um).

4- Enhances venous and lymphatic return (respiratory pump).

5- Synthesis of surfactant, Prostaglandins, histamine, and activation of angiotensin II = (metabolic or endocrine function).

6- Removal of certain substances from the circulation such as some prostaglandins, noradrenaline , serotonin and ACh.

7- Lungs contain strong fibrinolytic system.as it releases tissue plasminogen activator

8- Conducting zone is also important for humidification and conditioning air.

9- The nose serves as the organ of smell.

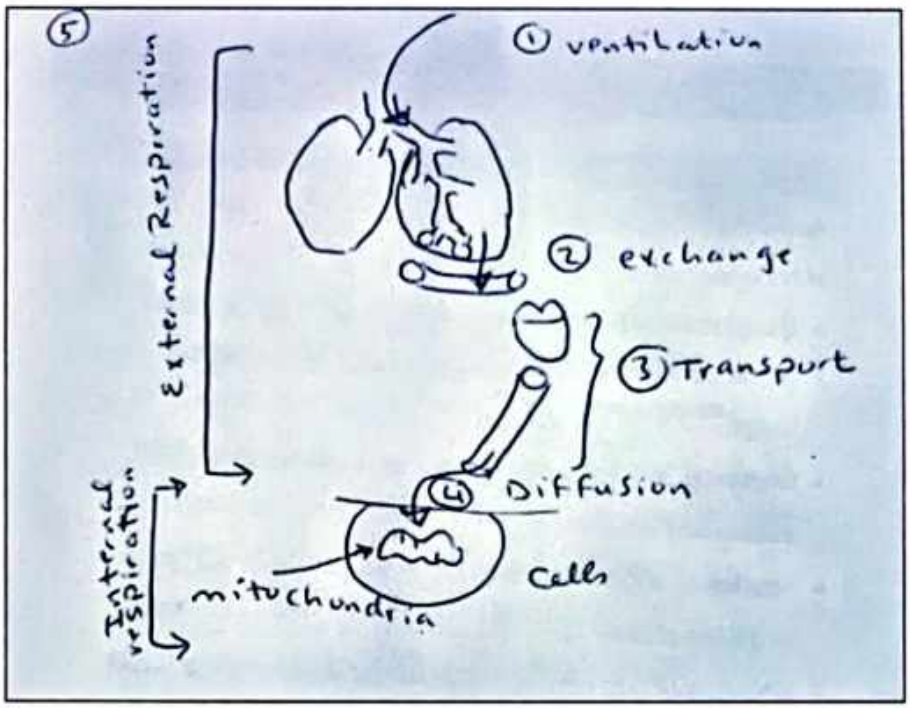
10- Vascular reservoir: pulmonary capillaries can store up to 500-1000 ml during exercise and deep inspiration. As the alveoli become recruited during exercise

Respiration composed of two processes:

External respiration: is the exchange of air between the body and external environment

It is composed of:

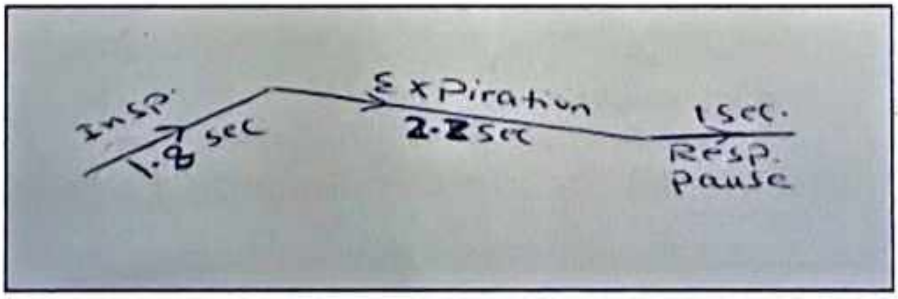
- **Ventilation**: is the exchange of air between atmosphere (external environment) and the air sacs (alveoli) of the lungs.
- **Exchange** of O2 and CO2 in the alveoli and blood within pulmonary capillaries.
- **Transport** of O2 and CO2 by the blood between the lungs and the tissues.
- **Diffusion** of O2 and CO2 between the tissue and the blood across the systemic tissue capillaries.



Internal respiration: is the intracellular metabolic process carried out within the mitochondria which use oxygen to drive energy (ATP) from food producing CO_2 as by-product.

Respiratory cycle is composed of:

- Inspiration (1.8 sec)
- Expiration (2.2 sec)
- Expiratory pause (1.0 sec) = midthoracic position
- Respiratory rate = 12 -16 breaths / min



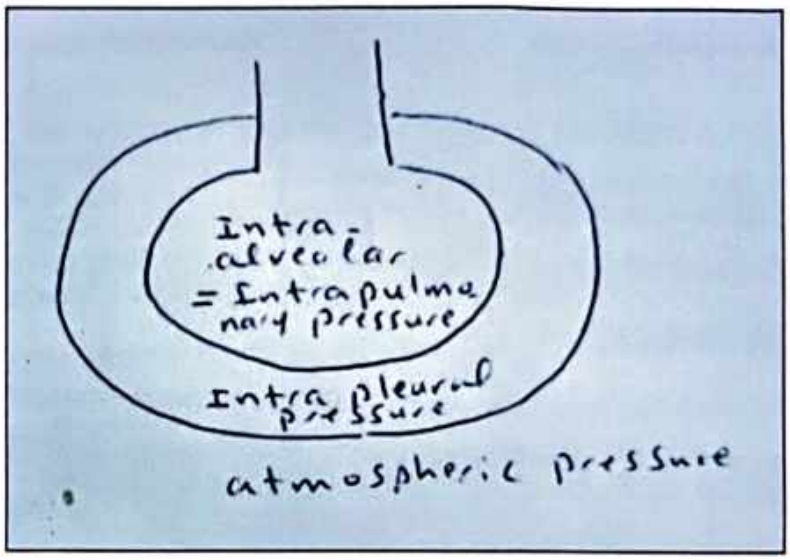
Important terms of respiration:

- **Eupnea:** easy rhythmic breathing during rest.
- **Apnea:** temporary stoppage of breathing.
- **Tachypnea:** increase in frequency without increase in the depth.
- **Dyspnea:** difficult breathing.
- **Orthopnea:** dyspnea only in recumbent position
- **Hyperventilation:** increased air flow to and from the lungs.
- **Hypoventilation:** decreased air flow to and from the lungs.
- **Hypercapnia:** increased CO2 in the blood due to hypoventilation.
- **Hypocapnia:** decreased CO2 in the blood due to hyperventilation.

Ventilation

- It is the mass movement of the air between atmosphere and alveoli. Air tends to move down pressure gradient.

There are 3 different pressures important in ventilation



1- Atmospheric (barometric) pressure: = 760 mmHg at sea level (Is considered to be 0 mmHg when relating other pressures to it).

It is decreased with increasing altitude above sea level.

2- Intra-alveolar pressure (intra pulmonary pressure)

- Is the pressure inside the alveoli
- Is less than atmospheric pressure by 1 mmHg during quiet inspiration (i.e.759 mmHg = -1 mmHg or negative pressure).
- Is greater than the atmospheric pressure by 1 mmHg during quiet expiration (i.e 761 mmHg = + 1 mmHg or positive pressure).

- It tends to equilibrate with atmospheric pressure due to direct continuity between the alveoli and the atmosphere

3- Intrapleural pressure (intrathoracic pressure)

- Is the pressure inside the plural sac.
- Is usually less than atmospheric pressure averaging about 756 mmHg or -4 mmHg in average during rest
- It doesn't equilibrate with atmospheric pressure or intra-alveolar pressure because there is no direct communication between them.
- Its negativity is important to inhibit lung collapse or chest hyperinflation.
- Its negativity is useful for suction of venous and lymphatic return against gravity.
- It is caused by recoil forces of elastic tissues present in lungs which tends to collapse and the chest wall which tends to hyperinflate. (they tend to pull from each other)
- Therefore, it can be used as an index and chest wall elasticity Can be measured by intra esophageal balloon

Transmural pressure

- It is the pressure difference between the inside and the outside of the membrane of cavitary structure

1- **Transpulmonary Pressure** (across the lung wall)

Intraalveolar pressure – intrapleural pressure

760 – 765 = +4 mmHg

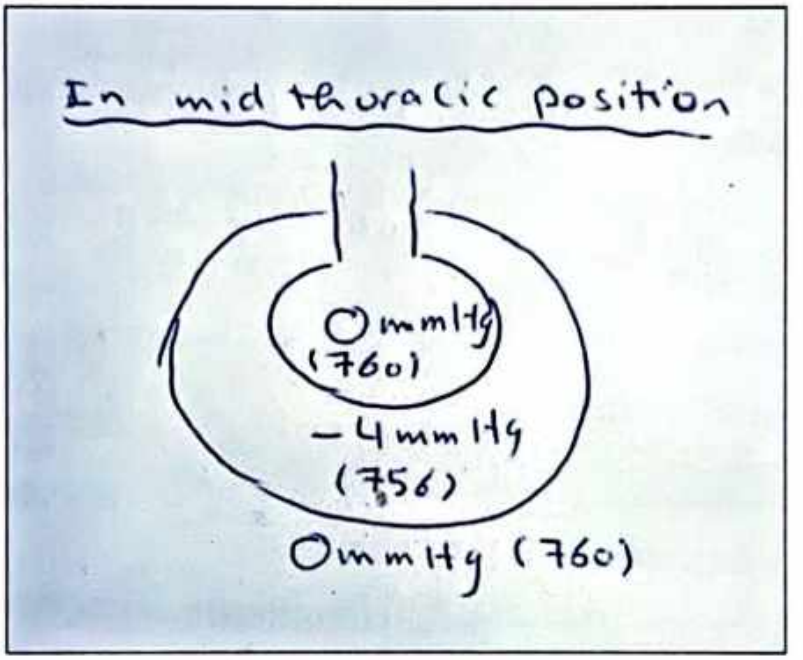
- This pressure should be positive throughout the respiratory cycle because the intraalveolar pressure (which inflate the lung alveoli) should be more than the intrapleural pressure (which compresses the lung alveoli) → therefore, the transpulmonary pressure is called distending pressure.

2- **Transthoracic pressure** (across thoracic wall)

Intrapleural pressure – atmospheric pressure = - 4 mmHg

- This pressure is negative because the atmospheric pressure that compresses the chest wall should be more than the pleural pressure which distend the chest wall,

this inhibits chest hyperinflation → compressing pressure.



NB: Intrapleural pressure IPP is

-6 mmHg at the end of normal quiet inspiration.

-4 mmHg at the end of normal quiet expiration.

-30 mmHg during deepest inspiration.

+30 mmHg during maximal forced expiration.

Pneumothorax

Definition : presence of air in the plural sac

Types:

1- Traumatic = open pneumothorax

Occurring due to injury of chest wall e.g.(stab wound)

2- Closed (spontaneous) pneumothorax

Occurs due to ruptured alveoli in pleural sac

- In both conditions , air enters from atmosphere into pleura → loss of negativity of intrapleural sac → lung collapse (atelectasis) & chest hyperinflation.

Mechanics of respiration

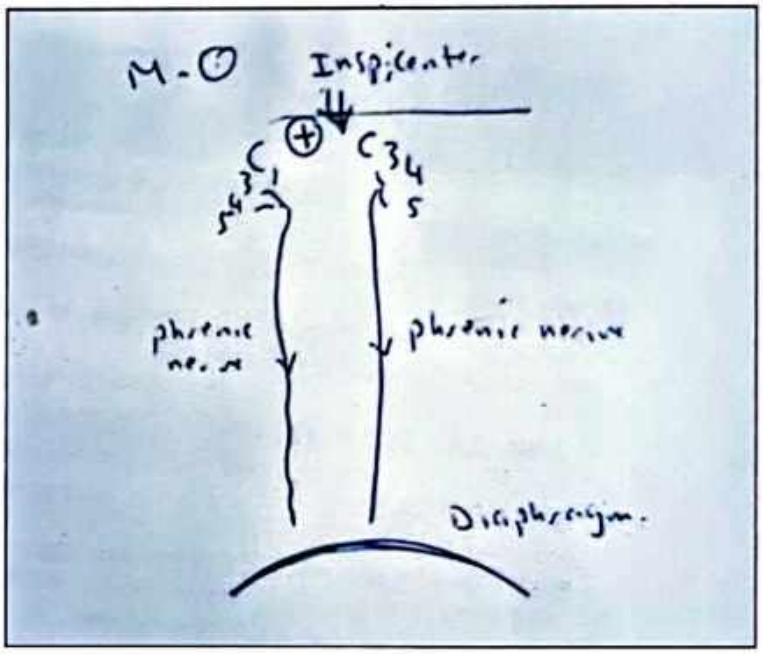
- **Inspiration:** movements of the air from atmosphere through airways into the alveoli.
- **Expiration:** movement of the air from alveoli through airways to the atmosphere.

Inspiratory muscles:

1) Diaphragm

- Is the major inspiratory muscle 75 %
- When contracts it moves down → ↑ vertical diameter of thoracic cavity.
- Each 1 Cm increase in vertical dimension leads to suction of 500 ml of air from atmosphere
- Paralysis of the diaphragm → respiratory paralysis (stop respiration) = apnea.

- Diaphragm is supplied by phrenic nerve C3,4,5
- Cutting of the spinal cord above cervical segment 3 → paralysis of the diaphragm → respiratory paralysis → death
- Cutting of the spinal cord below cervical segment 5 → the person is completely paralyzed (All the skeletal muscles ,except diaphragm, are paralyzed) → but the person is still able to breath because contraction of the diaphragm is the main respiratory muscle.

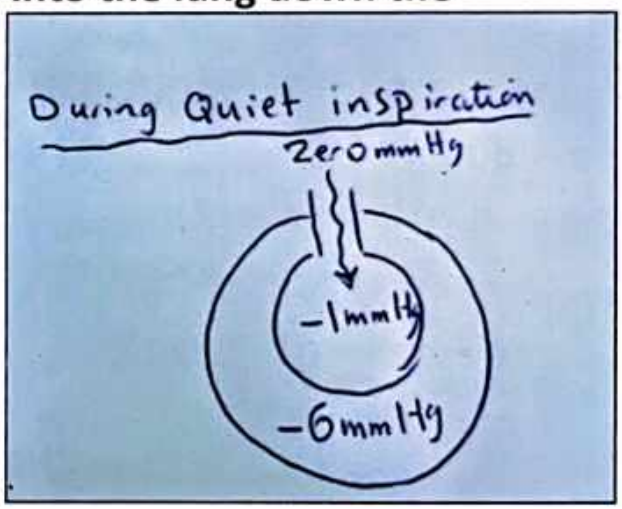


External intercostal muscles

When contract → ribs move upward and forward (bucket handle manner) → ↑ anteroposterior and transverse diameter of the thoracic cavity (eversion and elevation).

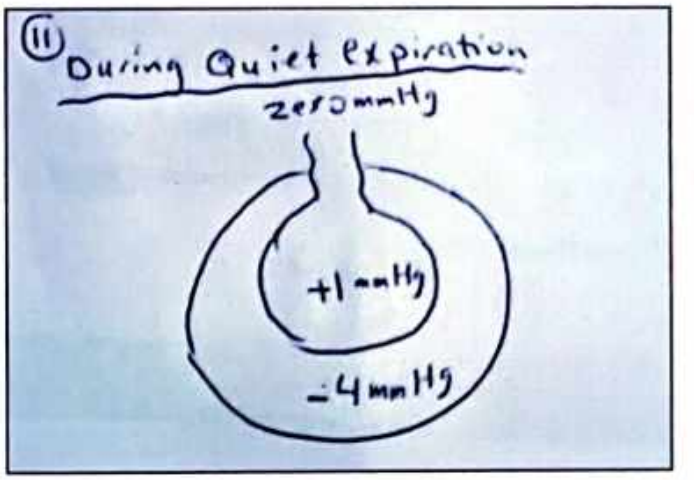
Mechanics of quiet(tidal) breathing:

- **tidal inspiration** starts by contraction of diaphragm and external intercostal muscles → the thoracic cavity enlarges →more negative IPP (- 6 mmHg) → increased transmural pressure across the lungs → the lungs are forced to expand to fill larger thoracic cavity → ↓interalveolar pressure by 1 mmHg below atmospheric pressure (-1 mmHg) → air flows into the lung down the pressure gradient , until intra-alveolar pressure equalizes with atmospheric pressure, this volume of air 500 ml.



- **Tidal expiration** starts when the diaphragm and external intercostal muscle relax → increased IPP to be (- 4 mmHg) again → the stretched lungs recoil (return) to their pre-inspiratory size because of its elastic properties (like release of a stretched balloon) → ↓ lung volume → ↑intra-alveolar pressure by about 1 mmHg above

atmospheric pressure (+1 mmHg) → air leaves the lung down its pressure gradient until intra-alveolar pressure becomes equal to atmospheric pressure, This volume of air 500 ml.



N.B

- The intra-alveolar pressure = 0 mmHg at midthoracic position and at the end of inspiration & expiration.
- Inspiration is always an active process, done by contraction of inspiratory muscles with energy utilization.
- Expiration is a passive process during quiet breathing since it is done by relaxation of inspiratory muscles with no muscular exertion and no energy expenditure.

Mechanics of deep breathing as during exercise:

A) Deep inspiration

- Strong contraction of the diaphragm and external intercostal muscles. (Diaphram moves 7 cm downwards)

- Contraction of the accessory inspiratory muscles (scalenus and sternocleidomastoid).
- This causes further increase in thoracic cavity → more negative IPP (-30 mmHg) → ↑ in lung volume than under resting conditions → more ↓ in intra alveolar pressure → more air enters the lung (lungs can inspire up to 3500 ml of air by deepest inspiration).

B) Deep (Forced) Expiration

- Is an active process
- Requires contraction of the expiratory muscles , abdominal muscles & internal intercostal muscles.
- Thus contraction of the expiratory muscles causes further reduction in thoracic cavity which leads to further reduction in lung volumes → ↑ intra-alveolar pressure more than during quiet expiration (may reach + 40 mmHg) → more air leaves the lung during forced active expiration than during quiet passive expiration.

Notice that :-

- Paralysis of intercostal muscles alone → does not seriously influence quiet breathing

Pulmonary surfactant

Definition: It is a substance that act to decrease surface tension of the alveolar membrane.

Origin: It is secreted by specific secreting cells (Type II Pneumocytes)

Chemistry: It is a complex mixture of :

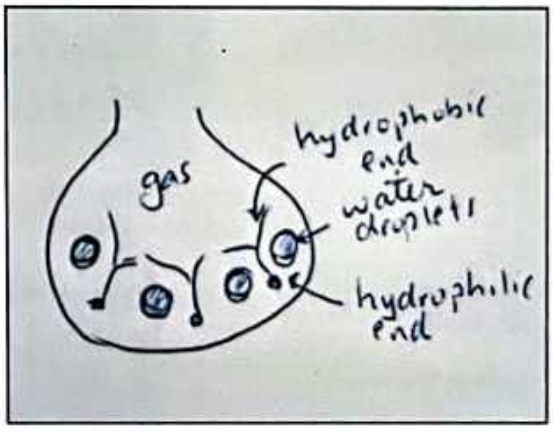
- phospholipids (lecithin)
- Apoproteins
- Calcium ions

WHAT IS MEANT BY ALVEOLAR SURFACE TENSION ?

- It is the attractive forces between the molecules in the fluid film that lines the alveoli which make alveoli resist inflation and become less in size.

Role (function) of surfactant:

1) Decrease the surface tension of the fluid lining the alveoli Because the surfactant has a hydrophilic end (water soluble) lie between fluid molecules and a hydrophobic end



(water insoluble) lie towards gas molecules.

Surfactant forms a layer between the fluid lining the alveoli and air in the alveoli

2) Keep alveoli dry and prevents pulmonary edema

By reduction of alveolar surface tension → decreasing filtration forces of fluids from pulmonary capillaries into alveoli → dry alveoli

3) Help to stabilize the size of the alveoli preventing of lung collapse:

Surfactant diminishing the surface tension in smaller alveoli than in larger alveoli , otherwise the small alveoli collapse and empty in large ones if surface tension were similar in large and small alveoli.

- According to laplace law , $P = \frac{2XT}{r}$.

  P= pressure, T = tension, r= Radius.

- Because the radius is low in small alveoli compared to lager alveoli, the surface tension of small alveoli should be reduced to inhibit the increase that might occur in their pressure NB: in small alveoli,(lower radius) → higher

pressure → collapse of these alveoli, but surfactant decreases tension → decreases pressure in small alveoli to become like that in large alveoli → no collapse.

4) Surfactant increases lung compliance during inspiration.

5) Surfactant prevents alveolar collapse during expiration.

6) Surfactant decreased work of breathing.

Surfactant deficiency occurs in:

Respiratory distress syndrome (RDS)

A) In premature babies (hyaline membrane disease)
Surfactant deficiency → ↑ surface tension → ↓ compliance → ↑ work of breathing → may lead to inspiratory failure.
↑ surface tension → ↑ flow of fluid from capillaries to alveoli → accumulation of plasma ultra filtrate fluid → glassy pale blue staining membrane lining in the alveoli (Hyaline membrane).

They may die at birth due to respiratory failure.

Treatment

- By positive pressure ventilation (PEEP)
- Administration of cortisol (glucocorticoids) to pregnant woman (who deliver premature) before birth.

NB: Cortisol hormone causes maturation of the surfactant and reach high levels only at full term.

- Inhalation of synthetic surfactant.

B) Adult respiratory distress syndrome:

Causes:

1- Hypoxia : which impairs synthesis of normal surfactant

2- Inhalation of 100 % O_2 for long time → ↓ synthesis of surfactant.

3- Occlusion of one pulmonary artery.

4- Smoking inhibits surfactant secretion.

5- Patient with cardiac surgery using pump oxygenation with interruption of pulmonary circulation.

6- Hypothyroidism.

7- Hypocortisolism (Cortisol accelerate maturation of surfactant).

Compliance

Def: The extent of lung expansion (change in volume) for each unit increase in transmural pressure.

Or, The change in lung volume (ΔV) per unit change in transmural pressure gradient (ΔP).

$C = \Delta V / \Delta P$

- It is a property of any vesico-elastic structure.

Vesico = having a lumen

Elastic = having elastic element

- Higher compliance = Easier expansion & Easier inflation at a given degree of transmural pressure gradient

Normally

- Compliance of lung = 0.2 L / Cm H_2O
- Compliance of lung + chest wall = 0.1 L/ Cm H_2O

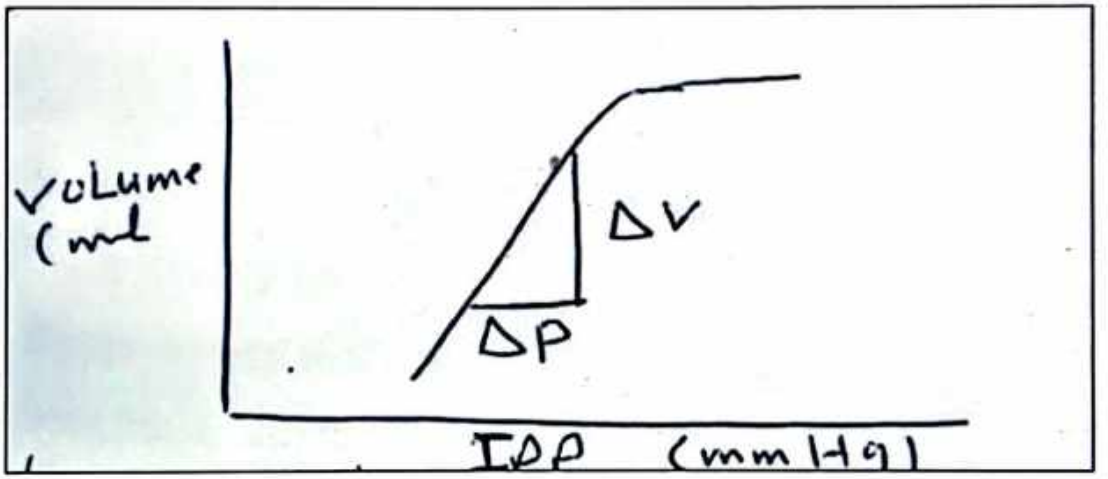
i.e, Lung expands easier than the whole respiratory system.

N.B

- Elastic elements resist changes in volume , therefore , a lung with increased elasticity **(high elastic recoil)** is harder to inflate. ↑ elastance → ↓Compliance.

Pressure volume curves

- C is calculated from steepest part of the curve.



Factors affecting compliance:

1- Initial lung volume

- Increased initial volume → ↑ Compliance

  So, compliance of adult lung is more than infant's lung

2- Surface tension

- ↑ surface tension **(stiff lung)** → ↓ compliance
- Surfactant (↓ S.tension) → ↑ compliance

3- Compliance is reduced in

a) Pulmonary congestion , edema

b) Pulmonary fibrosis

c) Respiratory distress syndrome (RDS)

Due to ↓ surfactant → ↑surface tension

d) Unilateral pneumonectomy (one lung)

Due to ↓ initial lung volume to the half of the normal person.

4- Compliance is increased in :

Old age, emphysema due to decreased elastic forces (↓ elastic recoil) → easier inspiration but no spontaneous expiration → the respiration becomes an active process.

Elastic behavior of the lung

1- Compliance important for lung expansion during inspiration.

2- Elastic recoil

- It is important after the end of inspiration to return lungs to pre-inspiratory volume (i.e. important during expiration).
- Elastic forces of the lung include

a. Lung elasticity → presents due to elastin & collagen fibers.

b. Alveolar surface tension → represents 2/3 of total elastic forces.

NB: Lung elastic recoil and alveolar surface tension favor alveolar collapse

Thus: lung expansion requires energy to

a- Stretch the elastic tissue of lungs
b- Overcome the surface tension of alveolar fluid.

NB:

Emphysema:

- A disease in which there is a decrease in α 1 antitrypsin → no inhibition of elastase enzyme → ↑ elastase activity → destruction of elastin protein → ↓ elastic recoil of alveoli → ↑ compliance.
- There is early collapse of small airways & difficult expiration.
- It is an obstructive airway disease (↑ airway resistance).

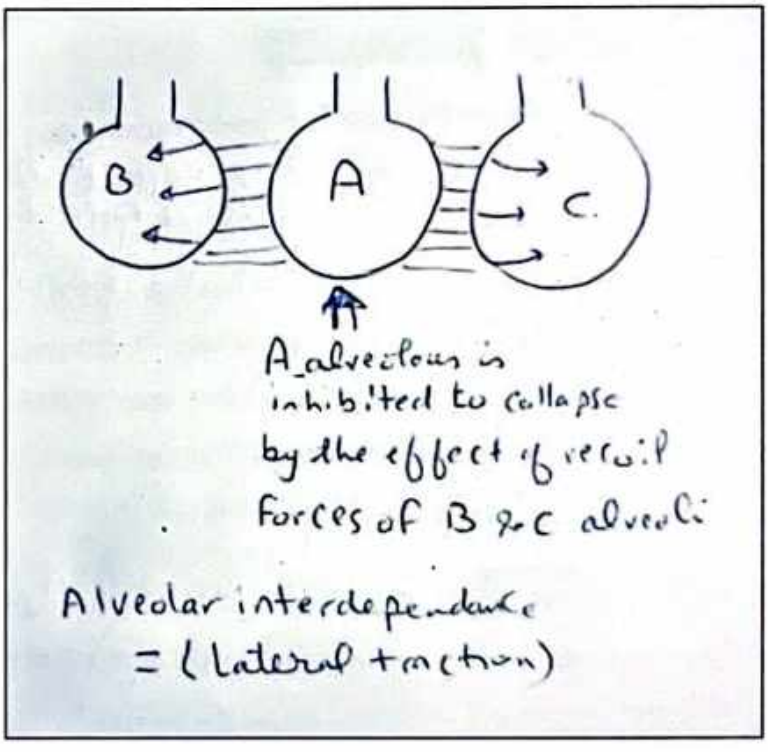
Factors that keep the alveoli open

1- Transmural pressure gradient (+ve transpulmonary pressure) I. alveolar P is > IPP at all conditions (even at forced expiration).

2- Surfactant (↓ Surface tension)

3- Alveolar interdependence= lateral traction

When an alveolus tends to collapse → interconnected alveoli become stretched and then they recoil and pull outward on the surrounding collapsing alveoli → force collapsing alveolus to open. Recoil forces of alveolar elastic tissue resist collapse at the interconnected surrounding alveoli.



Work of breathing:

- Work performed by respiratory muscles during breathing.
  - 3 % of total energy expenditure during rest.
  - 5 % of total energy expenditure during exercise.
  - 30 % of total energy expenditure in low compliance.

Types

1. Elastic work → 65 % → needed to overcome elastic recoil and increased in lung fibrosis e.g RDS
2. Non elastic work = Resistive work → 35 % → overcome airway resistance & inelastic tissue resistance,20 % (due to friction of lung tissues against each other during respiration).

Causes of increased work of breathing

1- ↓ pulmonary compliance

e.g Lung fibrosis & RDS.

2- ↓ elastic recoil → e.g Emphysema

That makes expiration active

3- ↑ airway resistance (e.g Bronchial asthma)

Air way Resistance

- Flow = ΔP/Resistance.
- If the resistance of airway is increased → the pressure gradient should be also increased to maintain normal air flow.
- Increasing pressure gradient between atmosphere and alveoli, needing higher work of respiratory muscles i.e.↑ work of breathing.
- Resistance (R) = $8 \times L \times \mu / \pi r^4$
  - -L = length
  - -μ = viscosity
  - -r = radius
- It is difficult to change Length or viscosity of airway or air sequentially, yet, length of airway may increase as in cases of inspiration from tubes, or in mouth breathers, and the viscosity of air may change as in case of changing humidity.
- The main factor affecting airway resistance is the radius of tracheo-bronchial tree.

NB: Air flow rate is the amount of air flow per unit of time that flows through a particular device. While, Air velocity is distance traveled per unit of time. Velocity is inversely related to CSA, So slowest air flow is at the alveoli (better gas exchange)

N.B: Because R is inversely proportionate to r4 , mild changes in radius → major changes in resistance.

Bronchoconstriction → ↑R

Bronchodilatation → ↓R

N.B: R is more in large airway (low total cross sectional area) compared to small airway (high total cross sectional area).

R is highest in medium sized airways.

Factors affecting airway resistance :

1- Neural control:

- Sympathetic stimulation→ Bronchodilatation (↓airway R)
- Vagal stimulation→ Broncho-constriction (↑airway R).

2- Local control:

- CO_2 (↑alveoli CO_2 → (BD)
  (↓alveoli CO_2 → (BC)
  a) This is important to adjust ventilation/perfusion ratio.

b) Area having higher blood flow in lungs → ↑CO_2 delivery to alveoli → BD

c) While area with decreased blood flow → ↓CO_2 → BC.

- Histamine:
  - Released from mast cells in allergic conditions such as bronchial asthma.
  - Causes BC.

3- Lateral traction on airway

- During inspiration → these is higher transmural Pr. gradient → ↑ traction on small airways in lungs → BD → ↓ airway resistance.
- While the lateral traction and transmural Pressure gradient is decreased during expiration → higher resistance.

Chronic Obstructive Pulmonary Disease (COPD):

- ↑ airway resistance due to bronchoconstriction

→ Pressure gradient should be increased to maintain normal air flow → higher work of muscles.

COPD occurs in:

1. Chronic Bronchitis
2. Bronchial asthma
3. Emphysema

- Resistance is more during expiration (Hard expiration).

Bronchial asthma:

- Allergic condition leading to release of histamine and BC on exposure to allergens such as perfumes or dust.
- The condition become worse if edema and thickening of mucosa of airway occurred, and if mucus secretion is increased.

Chronic Bronchitis:

- Long – term inflammatory condition of the lower respiratory airways causing narrowing of airways.

Emphysema:

- α1 antitrypsin deficiency → no inhibition of elastase → ↑elastase activity → Breakdown of elastin protein →

irreversible collapse of small airways and breakdown of alveolar walls.

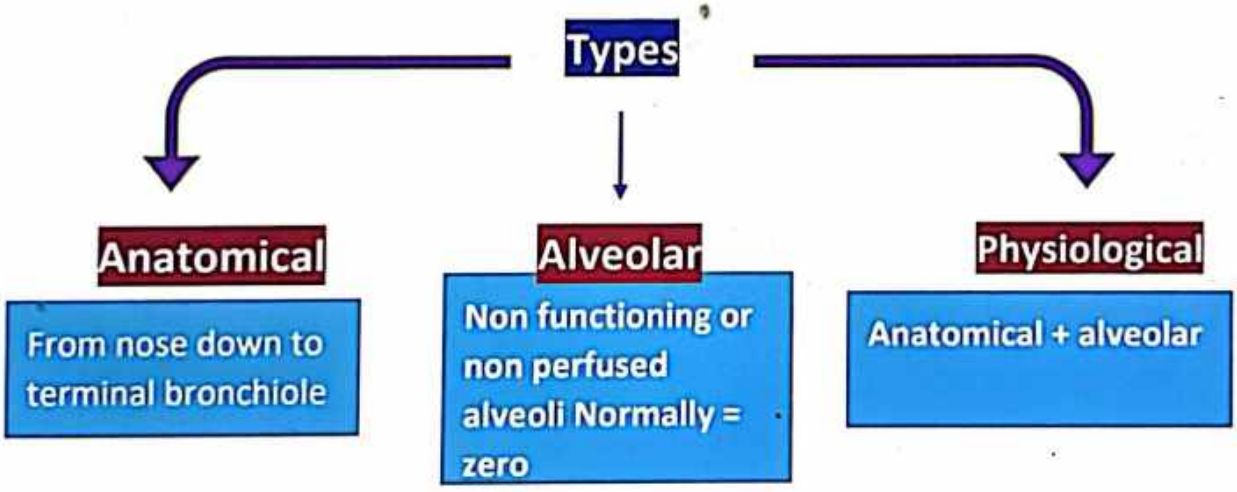
- Collapse of small airways cause increase in airway resistance.

The dead space

Def:

- Part of respiratory system which doesn't take part in gas exchange.
- Its volume = 150 ml

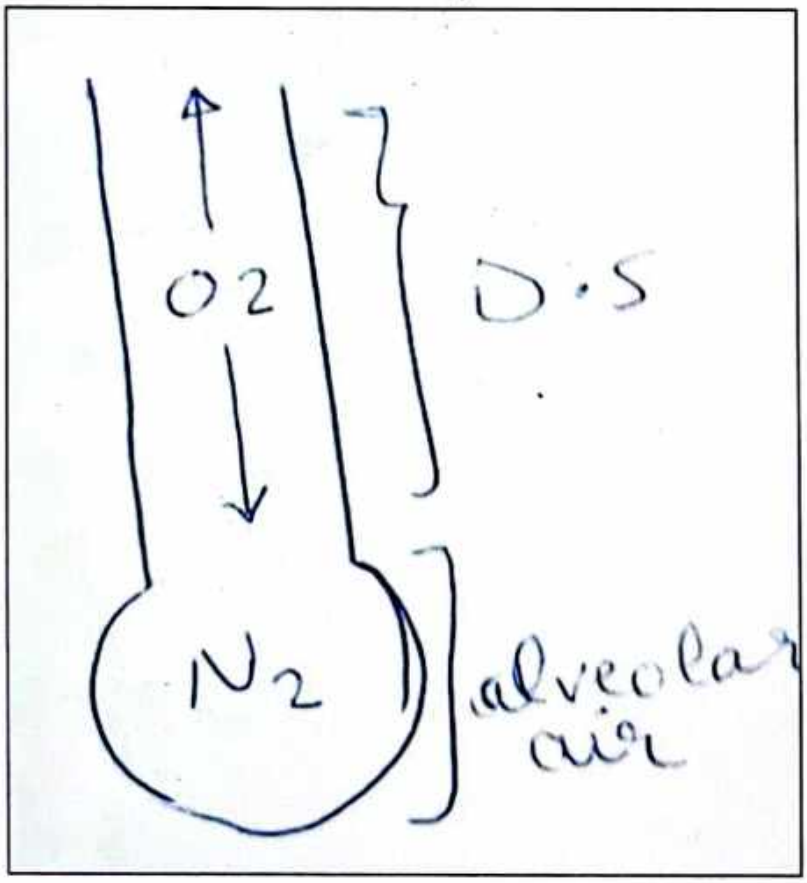
Types



- Normally all alveoli are functioning and perfused with blood , therefore , physiologically D.S = Anatomical D.S in normal person.

Measurements (Fowler's method) = nitrogen meter method

- Make the person inspires pure O_2 (100%).
- Then let him expires through a gasometer.
- Measure the gas (nitrogen) concentration in the expired air.
- The 1st expired air is O_2 (no nitrogen is determined)
- Once N_2 become recorded , the alveolar air is expired
- Dead space volume , is the volume of expired air until N_2 become recorded.



Significance of Dead space

1- Constitutes the conducting zone of the respiratory system which is important in (swellung , phonation)

2- Moisten & warm the inspired air.

3- Protection :

- 10 μ → hair of nose.
- 2-10 μ → mucous in the trachea , bronchi then expelled by cilia.

4- Prevent sudden changes in gas pressure of the alveolar air. (causes dilution of gases during inspiration).

5- Responsible for the difference between the total & effective pulmonary ventilation

e.g Total pulmonary ventilation = TV X RR

= 500 X 12 = 6L / min

Effective pulmonary ventilation = (TV-DS) X RR

= (500-150 =350) X 12 = 4.2 L/min

✓ In cases of rapid shallow breathing as in cases of pneumonia,

Total pulmonary ventilation = TV X RR = 200 X 30 = 6 L / MIN

Effective pulm. Ventilation = (TV – DS) X RR = 200 -150 X 30 = 1.5 L / min.

6- Responsible for the difference between alveolar air and expired air (expired air contains more O_2 , lesser CO_2).

Ventilation /perfusion ratio (V/Q ratio)

Def: Ratio of alveolar ventilation (V) to pulmonary capillary perfusion (Q)

Value: $\frac{4L}{5L}$ = 0.8

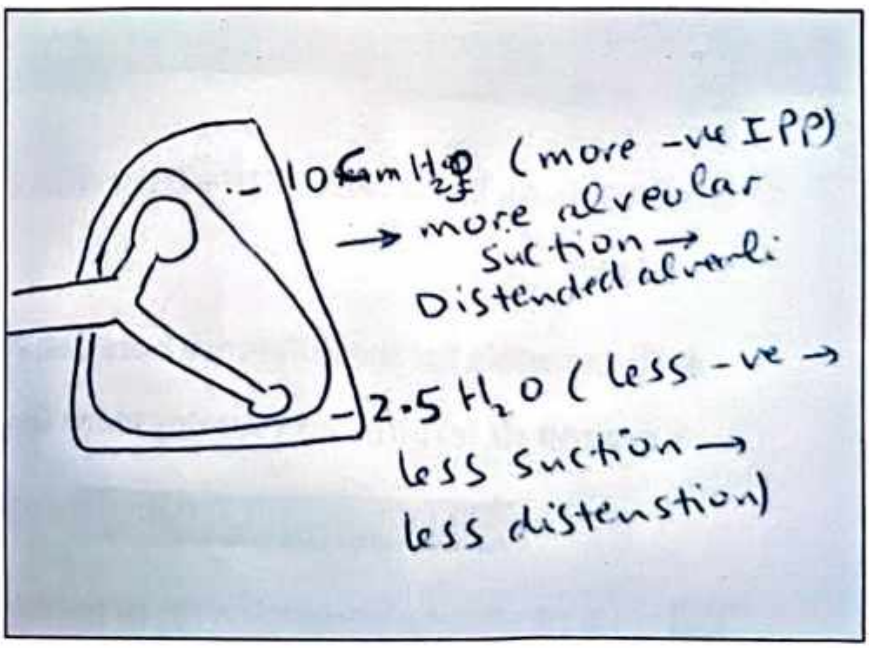
✓ This ratio is not constant in lungs

- It is higher at the apex = 3
- It is lower at the base = 0.6

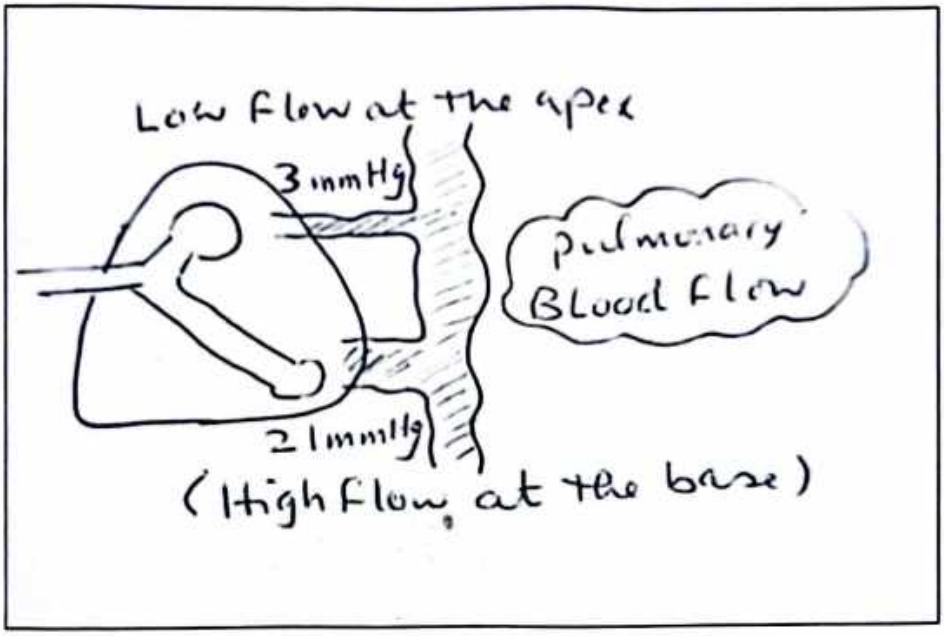
V/Q at the apex

- Alveoli at the apex are initially distended because of more -ve intra -pleural pressure than at the base.
- Distended alveoli at the apex are less compliant

  i.e, can not distend more for considerable value



- Therefore, (V) at the apex is low.
- Perfusion is also low at the apex because pressure Of blood vessels = 3mmHg above heart level (At the apex)



- Therefore,There is low (V), low perfusion (Q) at the apex but perfusion is much (markedly) decreased compared to ventilation. i.e $\frac{\text{Low ventilation}}{\text{Very low perfusion}} \rightarrow$ V/Q is high at the apex = 3

V/Q at the base

- (V) at the base is high (more compliant alveoli) because of these initial low volume → more expandable.
- Perfusion Q at the base is high because the pulmonary vessels pressure at the base = 21 mmHg.

- $\frac{\text{high ventilation}}{\text{Very high perfusion}}$ → low V/Q = 0.6.

N.B

- value of V/Q mentioned before are applied on standing , setting position (gravity factor)
- On recumbent position, V/Q becomes higher on the anterior parts of lungs, and low at the back of the lungs.

N.B most efficient V/Q for proper gas exchange = 1

Significance of V/Q

1- V/Q higher than normal

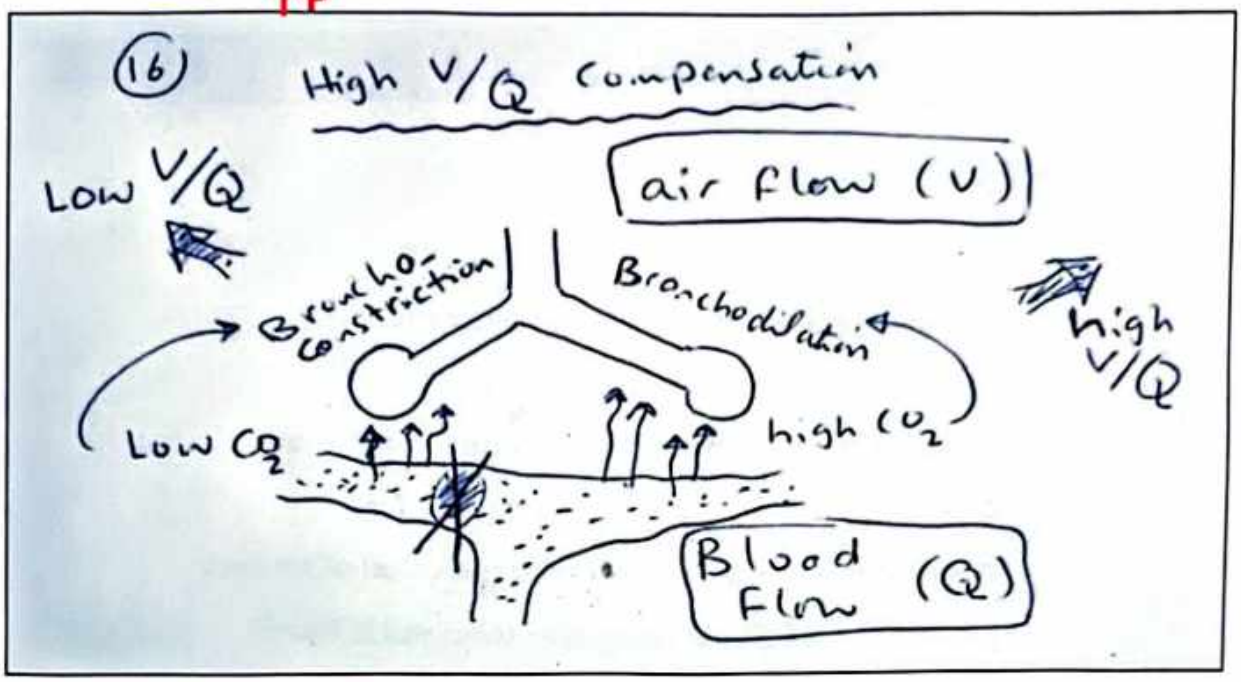
- Causes increased physiological dead space

  i.e , air in the non perfused alveoli.
- More air than blood is called wasted ventilation
- Correction (homeostatic mechanism):

  When there is a cause for underperfusion (low Q) of an area (alveoli), the CO_2 exchanged in this area is low → pulmonary broncho-constriction → low V

  - $\frac{\text{Low V}}{\text{Low Q}}$ → normal ratio.

- While the well perfused alveoli have high concentration of CO_2 → Bronchodilation → ↑ V
  - i.e, $\frac{\uparrow V}{\uparrow P}$ → normal ratio.



2- V/Q lower than normal

- This causes physiologic shunt
  - Shunted blood = blood that leaves the lung without oxygenation.

✓ It includes :

- 2% of COP passing through bronchial vessels, not through pulmonary vessels.
- Low V/Q = inadequate ventilation

- Because, inadequate ventilation leads to passage of venous blood through pulmonary capillaries without oxygenation.

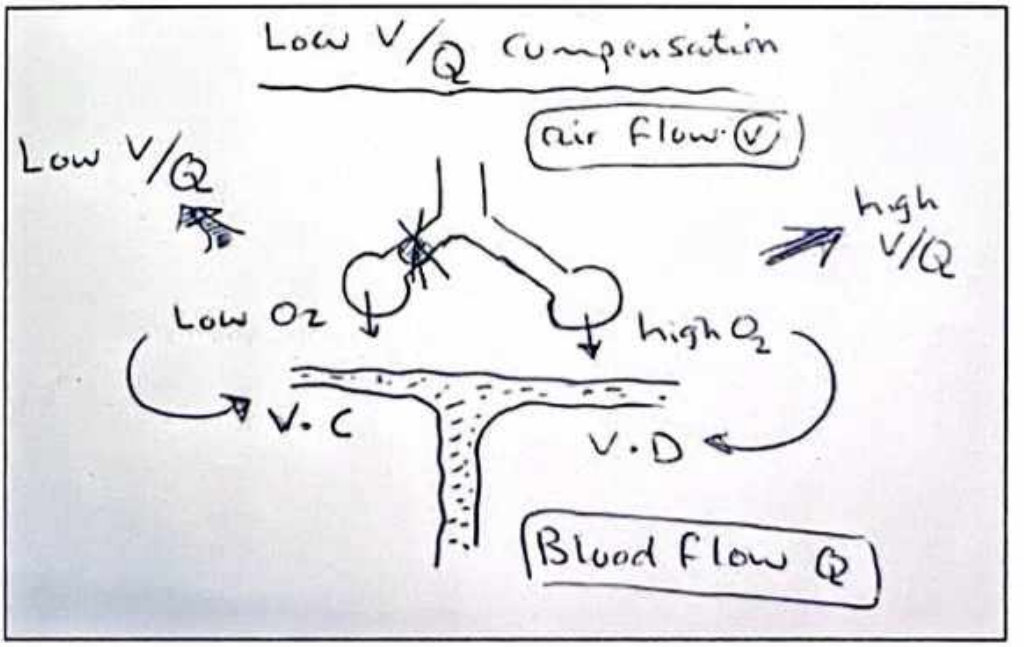
- Correction (homeostatic mechanism)
  - If there is a cause for underventilation of some alveoli e.g (mucus plug) → hypoxia in these alveoli causes vasoconstriction → ↓ Q

i.e, $\frac{\downarrow V}{\downarrow P}$ = normal ratio.

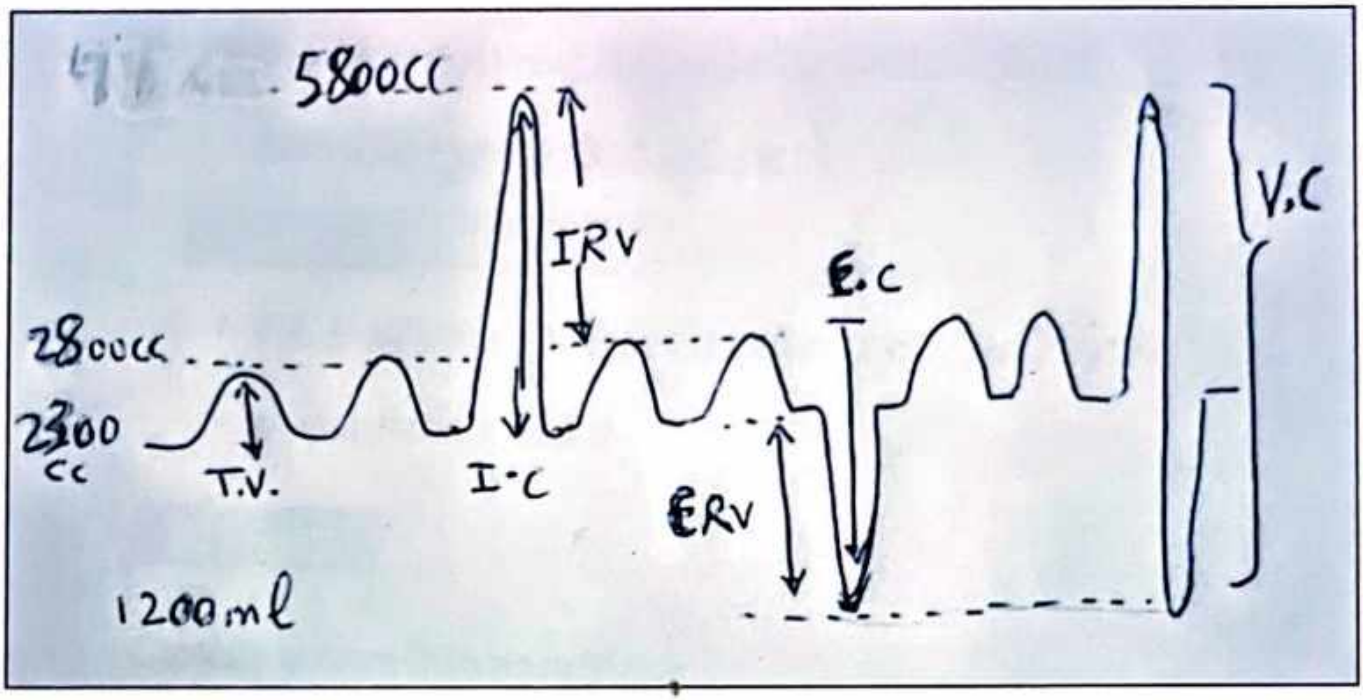
  - While, well ventilated alveoli have high O2 concentration → vasodilation → ↑ O2

i.e, $\frac{\text{High V}}{\text{High Q}}$ → normal ratio.

N.B hypoxia → V.C in pulmonary vessels & VD in systemic vessels.



Lung volumes & capacities (Practical)



Lung volumes

1- Tidal volume T.V = 500 ml

- Vol. of air which could be inspired or expired during rest.

2- Inspiratory reserve volume IRV=3000 ml

- Vol. of air inspired by maximal forced inspiration after normal tidal inspiration

3- Expiratory reserve volume ERV = 1100 ml

- Vol. of air expired by maximal forced expiration after normal tidal expiration.

4- Residual volume RV = 1200 ml

- Vol. of air remaining in alveoli after maximal forced expiration to inhibit lung collapse

5- Minimal air volume = 150 ml

- Vol. of air present in tissues even after opening of chest by stab wound and loss od R.V

Lung capacities

Capacity is more than one volume

I.C (inspiratory capacity vol. of air inspired by deepest inspiration after normal tidal expiration.

TV + IRV = 3500 ml

E.C (expiratory capacity) vol. of air expired by maximal forced expiration after normal tidal inspiration.

TV + ERV = 1600 ml

3- V.C (vital capacity) vol. of air expired by maximal forced expiration after deepest inspiration.

IRV + TV +ERV = 4600

4- FRC (functional residual capacity)

Vol of air remaining in lung after normal tidal expiration

ERV + RV = 2300 ml

5- **TLC (total lung capacity)** vol. of lungs after deepest inspiration.

V.C + RV = 5800 ml

Q:- How can Lung volumes & capacities be measured?

- Using SPIROMETER which draws a tracing called spirogram.
- It can measure any volume or capacity, but can not measure R.V or any capacity contains RV such as FRC & TLC.

So it can measure:

3 lung volumes	&	3 lung capacities
▪ TV		▪ IC
▪ IRV		▪ EV
▪ ERV		▪ VC

Q:- How can you calculate any of these volumes or capacities using spirogram ?

- The volume of one large square = 300 ml of air
- As the large square = 10 small squares
  so any small square = 300/10 = 30 ml
- For example: for measuring of TV in this

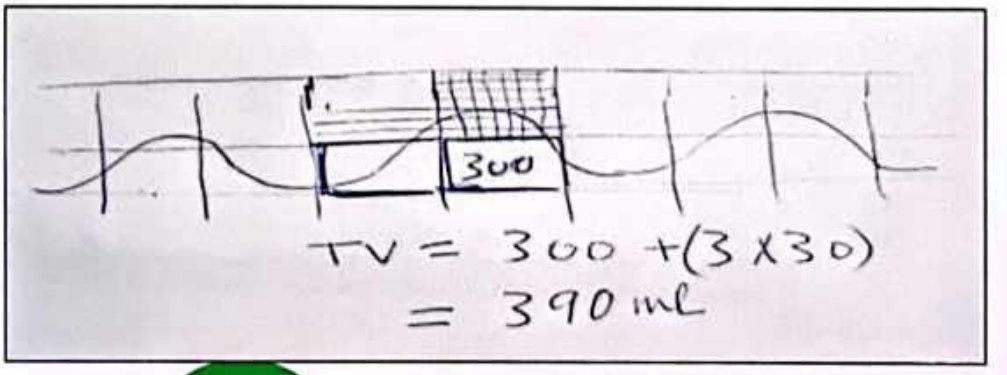


diagram → 300 + (3 x 30) = 390 ml

Q:- Can the residual volume be measured using the spirometer ?

- No, but it can be measured using Helium gas by a method depending on (dilution technique) and called Helium wash out technique.

Q:- Why the I.C is more than the E.C ?

- Because inspiration is an active process needing a center & (inspiratory muscles) while expiration in a passive process,
- Although the forced expiration is an active process, but muscles used during forced expiration are weaker than those of forced inspiration.

Q:- What is the importance of FRC & RV and minimal air volume ?

F.R.C: important to allow gas exchange between alveoli & pulmonary vessels even in the mid thoracic position (where no inspiration or expiration occurs).

This exchange inhibits fluctuation of body gases.

R.V:

1- Inhibit lung collapse.

2- Keep the alveoli compliant (easy to increase in volume) because compliance is more if the initial volume of alveoli is high, so in complete collapse → very low compliance.

3- Allow gas exchange even after maximal expiration.

Minimal air:

It is important in medico-legal aspects.

e.g to differentiate if the infant is born then died or died then born.

- Take a piece of lung and put it in a saline:
  - ✓ If it is lighter than saline (containing minimal air) → floating of the piece over the saline so the infant is born then died.
  - ✓ If it sinks → no minimal air vol→ (died then born).

Q:- What is the most important static test to asses lung function?

- V.C

Q:- What are the causes that ↓ TLC & VC?

1- Pulmonary edema & pneumonia

2- Lung fibrosis

3- Pleural effusion

4- Deformity of chest wall such as kyphosis & scoliosis

5- Pressure from downwards such an ascites, pregnancy.

Q:- What are the causes of ↑ TLC?

Emphysema (breakdown of the septa between alveoli & destruction of elastic tissue fibers due to increase elastase activity) This occurs when the α 1 antitrypsin is inactivated as in smoking 98 % / congenital 2 %

so the lung are inflated more than normal → barrel chest

- **in this case the expiration becomes an active process as there is no elastic recoil of the lung alveoli**
- **in emphysema →**
  - **↑ TLC**
  - **↑ LUNG COMPLIANCE**
  - **↑ FRC & ↑ RV**

Q:- Is emphysema obstructive or restrictive lung disease?

- **It is both obstructive & restrictive illness but mainly obstructive.**

Dynamic lung function

Dynamic means (relative to time)

1/ Respiratory minute volume (RMV)

- TV × Respiratory Rate

  1\2L × (12 – 16) = 6-8 L /minute

- It represents the Total Pulmonary Ventilation
  i.e, all air entering the chest/minute.

Q:- What is total & effective Pulmonary Ventilation?

- Total pulmonary ventilation:- 500ml × 12 = 6000ml/ minute.
- Effective pulmonary ventilation:- (500 – 150) × 12 = 4200ml / minute.

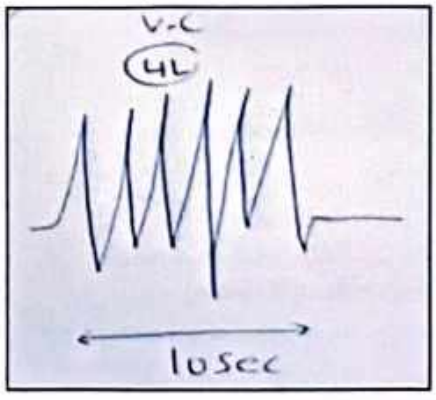
  i.e,= (TV – Dead space) × RR

  Effective ➡ is the volume of air in alveoli that enter the process of gas exchange.

2/ Maximum Breathing Capacity (MBC)

=Maximal Voluntary Ventilation (MVV)

- Maximal vol. of air ventilated by fastest & deepest Respiration per minute
- normally = 120 – 170 L/minute. (in males)



- Calculation:
  - measure one average V.C volume. e.g: 4 liters
  - VC × number of cycles / 10 sec.
    E.g: 4 × 6 = 24 Litre / 10 sec.
  - 24 × 6 = 144 litre /minute.

Q:- why we do this test in 10 sec. Only and not in complete minute?

- To avoid wash out of CO_2 ➡ Alkalosis➡ Syncopal attack اغماء

Q:- What are causes of decreased MVV ?

- Any lung, pleural or chest disease.

Q:- What is the most imp. Lung function TEST?

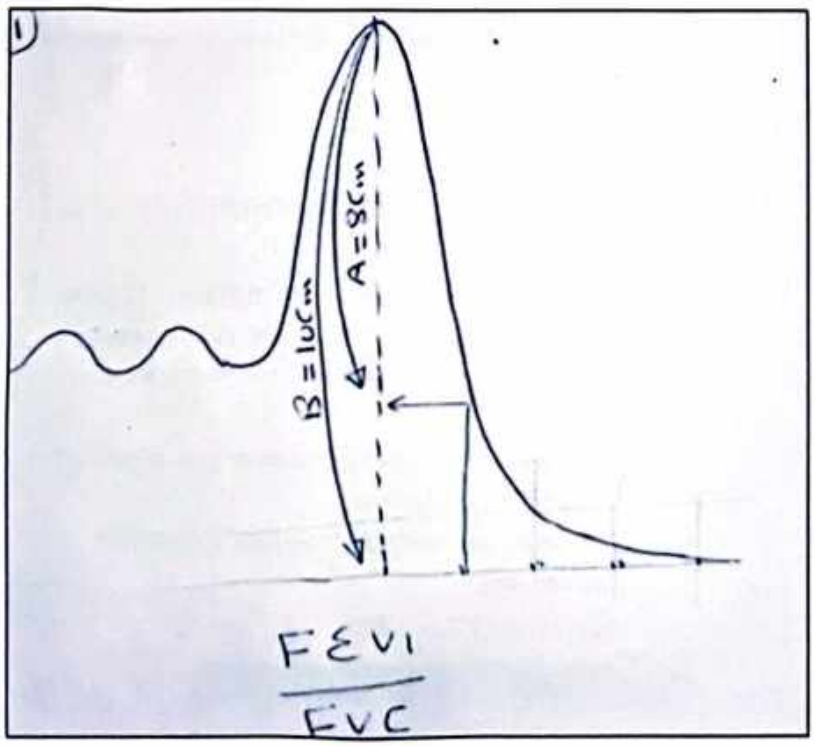
- V.C ➡ in static lung function.
- MVV ➡ in Dynamic lung function. (It is the most important)

Q:- What is the Breathing Reserve and Dyspneic Index ?

- BR= MVV (120 – 170) – RMV (6 – 8) e.g 130-8=122L
- DI = (MVV – RMV) \ MVV
  Normally > 90%
  If < 70% ➡ Severe dyspnea

3\ Timed vital capacity (FEV1)

- Volume of expired air during 1st sec. of maximal forced expiration.
- Normally 80%
- Calculation:-
  - Inspiration maximally followed by expiration maximally.
  - Draw a perpendicular line from peak of curve
  - After one large square (represents one sec.)
  - Draw another perpendicular line to cut the curve.
  - Again draw a 3rd perpendicular line to cut the 1st perpendicular line.
  - Measure the length of the line A over the whole line B
  - E.g:- 8cm\10cm = 80%



Q:- What is the Significance of this TEST ?

- FEV1 is decreased in Obstructive lung disease.
- Such as Bronchial asthma & Emphysema.
- Because the patient can't expire >80% of air in the obstructed air way in the 1st second.

Q:- what occurs in this Ratio in Restrictive Lung disease such as pulmonary fibrosis?

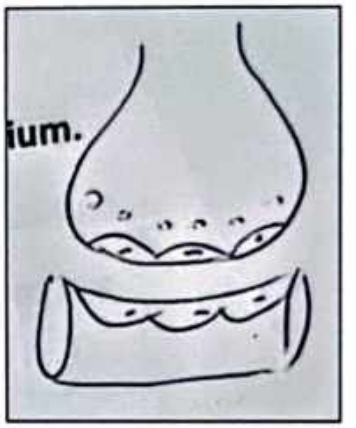
- The Ratio is unchanged in these diseases but the vital capacity is decreased as mentioned before.

	Obstructive Lung disease	Restrictive Lung disease
• Example	Emphysema & Bronchial asthma	Lung Fibrosis & Thoracic cage deformity
• Resp. disturbance	Expiratory	Inspiratory
• Lung Compliance	Increase (Emphysema)	Decrease or Normal
• Air way Resistance	• Increase • More difficult in expiration	• normal • More difficult in inspiration
• TLC	Increase	Decrease
• RV & FRC	Increase	Normal
• FVC	Decrease	Decrease
• FEV1	Markedly decrease	Decrease
• FEV1\FVC	Less than 80%	≥80%

Diffusion Of Gases

- Gas exchange between alveolar air and blood found in pulmonary capillaries
- occurs by simple diffusion across the Alveo-Capillary membrane which is formed of the following layers:

1. Thin fluid lining of alveoli.
2. Epithelial cells of alveoli.
3. Basement memb. of alveolar epithelium.
4. Basement memb. of capillaries.
5. Capillary endothelial cells.



Characters of thin membrane :

- Surface area = 70 .
- Thickness = 0.2 : 0.6 µM (average = 0.5 µM)

Factors affecting Diffusion of gases across the alveo-Capillary membrane:

(Fick's Law)

- Rate of gas diffusion = AD (P1 – P2) \ T
- A = Surface area of the filtering membrane.
- (P1 – P2) = Pressure gradient between alveolar air and blood.

- D = Diffusion co-efficient of gas = $\text{Soluibility} / \sqrt{\text{MW}}$

  So if gas is more soluble & less M.W ➡ it becomes more diffusible.

- T = Thickness of the membrane.

So, Diffusion will be affected be the following factors :

1) Surface area ➡ Direct relation.
   - ↑ SA (as in exercise ➡ ↑ Rate of gas diffusion.
   - ↓ SA (as in emphysema) ➡ ↓ Rate of gas diffusion.

2) Pressure gradient ➡ Direct relation.
   - ↑ Pr. gradient (as in exercise) ➡ ↑ Rate of gas diffusion.
   - ↓ Pr. gradient (as in emphysema) ➡ ↓ Rate of gas diffusion.

3) Thickness of membrane ➡ Inverse relation.
   - ↑ Thickness (as in pulmonary edema) ➡ ↓ Rate of gas diffusion.
   - ↓ Thickness (as in exercise) ➡ ↑ Rate of gas exchange.

4) Diffusion co-efficient

- Which depend on the solubility of gas in the respiratory membrane & its molecular weight.

N.B:

- Although the M.W of O_2 is less than that of CO_2 , Yet , The Diffusion ability of CO_2 is >20 times than that of O_2.
- This is because of much higher solubility of CO_2 compared to O_2 solubility (solubility co-efficient of CO_2 is 24 times that O2)
- So, Diseases which cause diffusion difficulties as pneumonia, Pulmonary edema (↑ Thickness of the membrane) will affect the diffusion of O2 (↓ its level in arterial blood). While CO_2 diffusion may remain normal and becomes affected only in severe cases.

Diffusion capacity of the lung:

- Diffusion capacity of lung to O_2 → 20 ml/min/mmHg.
- Diffusion capacity of lung to CO_2 → 400 ml/min/mmHg.

O2 Transport

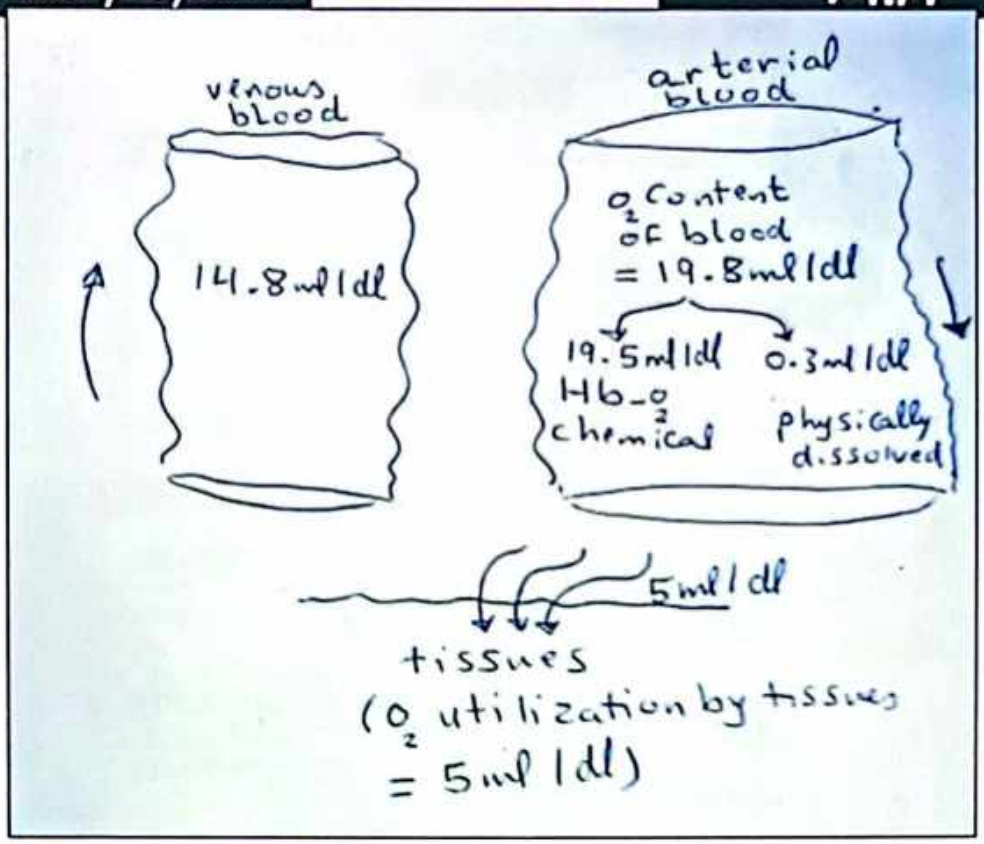
- O_2 is transported in the blood (from lungs to tissues) in 2 forms:

Chemical	Physical
(Hb- O2)	Dissolved in fluid
98.5% = 19.5 ml/ml blood	1.5% = 0.3 ml/dl
It is oxygenation, not oxidation	Directly proportionate to partial Pr. of O_2 (PO_2)
Each molecule of Hb can bind to 4 O_2 molecules.	Important For detection of O_2 flow according to Pr. gradient.

- Each one gram of Hb can bind to 1.34 ml of O_2

  Therefore, in 100 ml of arterial blood which contains 15 gm of Hb, about 20 ml of O_2 could be carried

  1.34 × 15 = 20 ml%.



-O_2 Content of Hb

- **Vol. of O_2 that are carried by Hb in 100 ml of blood = 19.5ml.**

-O_2 Carrying capacity

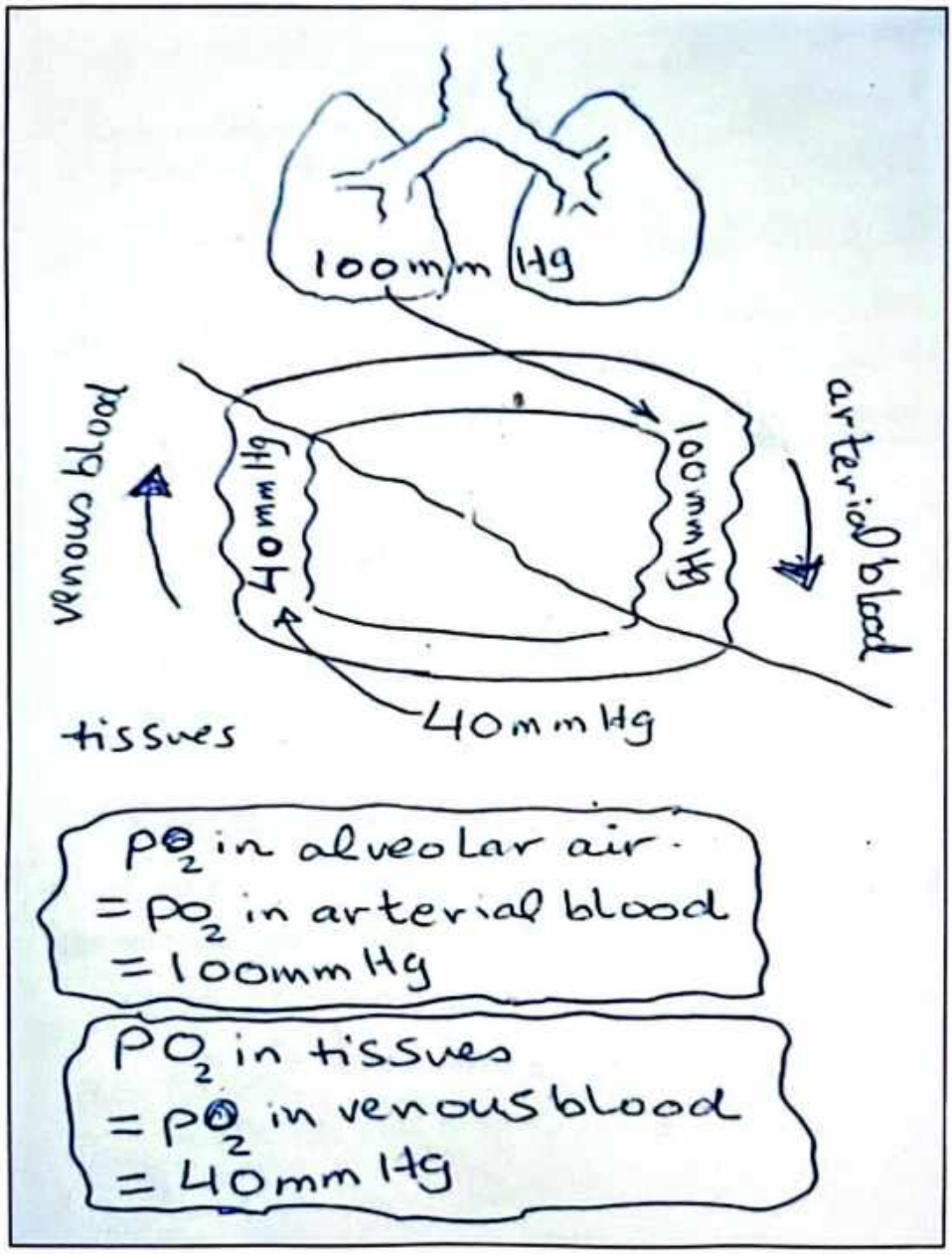
- **Vol. of O_2 that could be maximally carried by Hb in 100 ml arterial blood = 20 ml.**

-Percentage saturation

- **O_2 content/O_2 carrying capacity = 19.5/20 = 97.5%**

N.B: About 2.5% of Hb are in the met Hb form ie (oxidated Hb), and can't carry O_2 .

- So, the % of saturation is maximally 97.5% only ie, 97.5% only of Hb is saturated with O_2.
- PO_2 in alveolar air = PO_2 in arterial blood = 100 mmHg.
- PO_2 at tissue level = PO_2 venous blood = 40 mmHg.

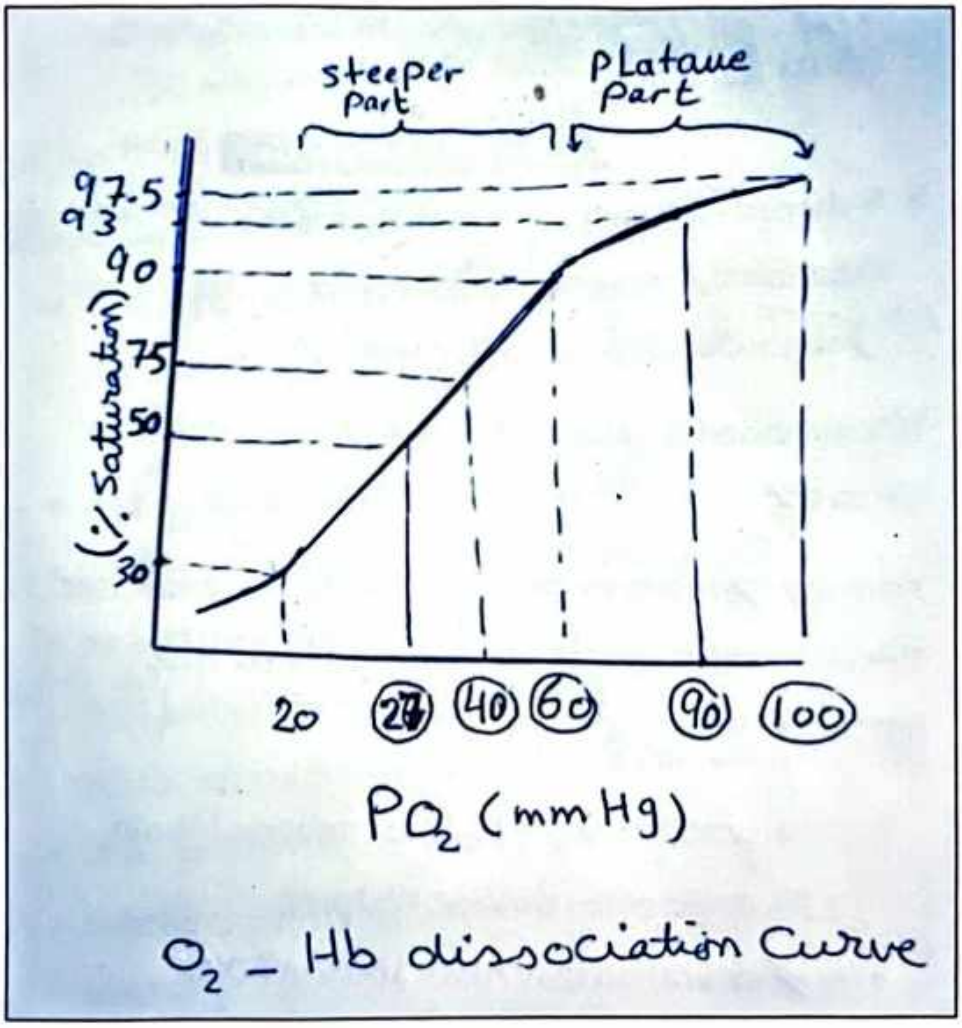


O_2 – Hb dissociation curve

- S-shaped (sigmoid) curve that studies the relationship between PO_2 (Partial Pr. of O_2) and percentage of saturation of Hb. (%saturation)

NB: Increased % saturation → Increased affinity of Hb to O_2

As every O_2 molecule that combines to Hb, stimulates the attachment of another O_2 molecule to Hb.



Plateau part of the curve:

- At PO_2 100 mmHg → 97.5% saturation (About 2.5% of Hb are in the met Hb form).
- % of saturation may reach 100% in PO_2 > 100 mmHg.
- At PO_2 90 mmHg → 93% saturation.
- At PO_2 60 mmHg → 90% saturation.
- This part of the curve shows that a major decrease in PO_2 is associated with a minor decrease in % saturation and affinity of Hb to O_2.

Significance:

- This gives a protective margin against hypoxia (ie, in hypoxic condition, where PO_2 is decreased markedly, Hb is still having high affinity to O_2 , and having high % saturation).
- This plateau part of the curve is applied at the lungs, where there is high PO_2 .

Linear part of the curve (steeper)

- Below PO_2 60 mmHg, the decrease in PO_2 is associated with marked decrease in % of saturation and affinity.
- At PO_2 40 mmHg → 75% saturation.
- At PO_2 27 mmHg → 50% saturation.
- At PO_2 20 mmHg → 30% saturation.

Significance:

- This part of the curve is applied at the tissue levels, where PO_2 is low.
- The rapid descent in % saturation and affinity is helpful to satisfy tissues with O_2 at low PO_2 levels.

  ↓ affinity → more delivery of O_2 to tissues.

P50 :

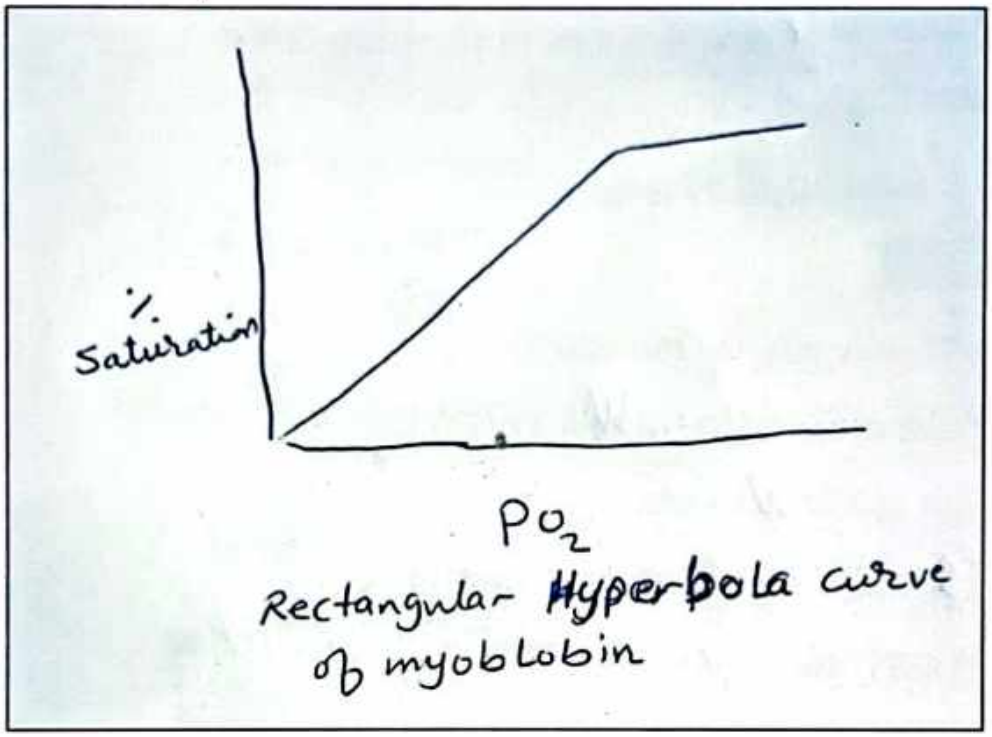
- It is the PO_2 at which 50% of Hb is saturated with O_2.
- It is normally = 27 mmHg.

Myoglobin:

- can carry one O_2 molecule.
- Molecule weight (MW) = 17,000 Dalton.

Ie: = 1/4 M.W of Hb.

- It's curve is rectangular hyperbola.
- At very low $PO_2 < 20$ mmHg → releases O_2 to the muscle (Myoglobin become the source of O_2 not Hb).

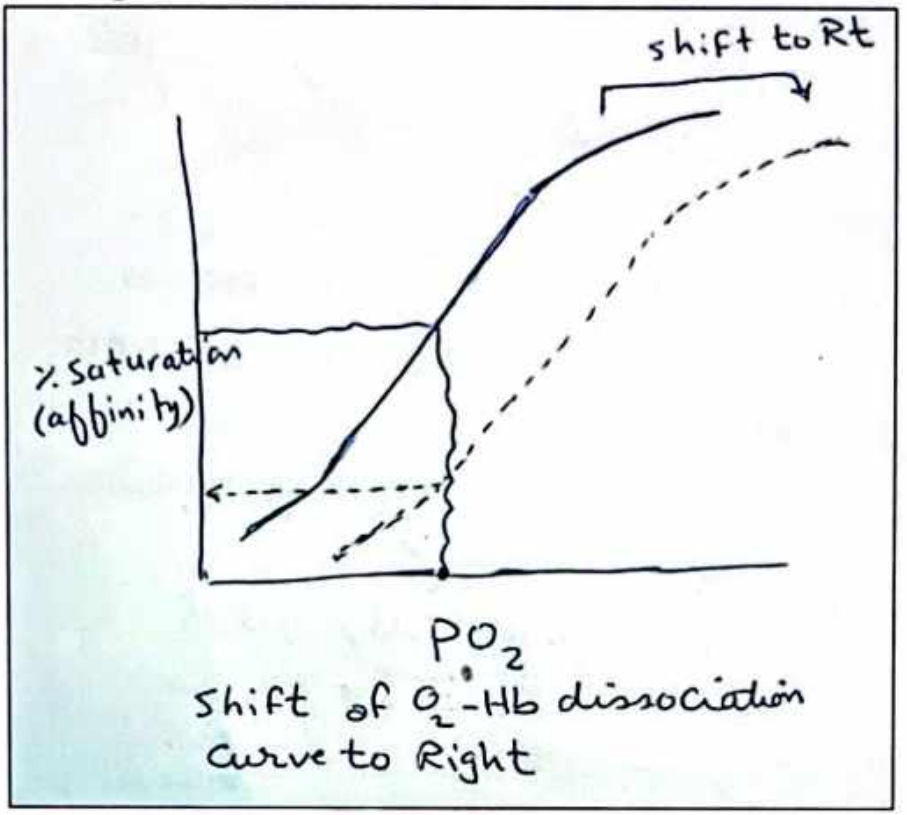


Shift of O2 – Hb dissociation curve

1- Shift to right:

- At any PO_2 , % of saturation (affinity) becomes lower when oxy-Hb dissociation curve is shifted to the right.

- Decreased affinity means that Hb supplies tissues with higher amounts of O_2.



Causes of shift to Right:

1- ↑H+ (acidosis) → Bohr's effect

2- ↑PCO_2 (hypercapnia) → Bohr's effect

3- ↑ Temperature

4- Polycythemia

5- ↑ 2,3 DPG

N.B:

2,3 DPG

- Product of glycolysis inside RBCs.
- Competes with O_2 for Hb (bind to β chain)

$$HbO_2 + 2,3\ DPG \rightarrow Hb\ 2,3\ DPG + O_2 \downarrow$$

(liberated to tissues)

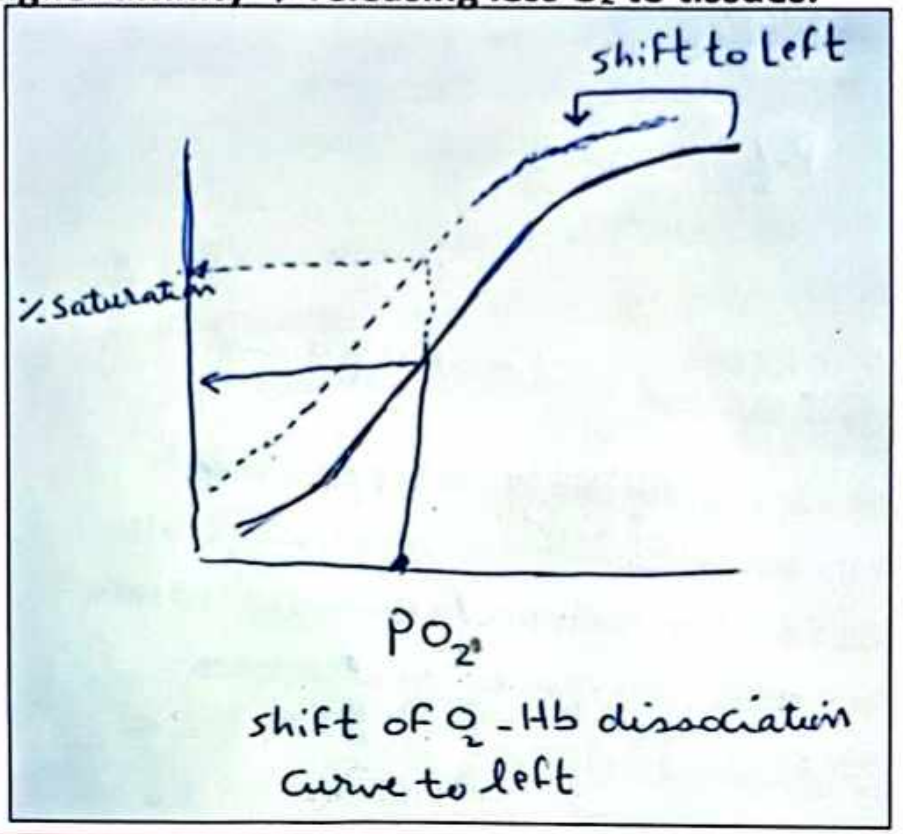
- ↑ in some cases of shift to Lt to antagonist their effect causing shift to Rt again.

  e.g: * Alkalosis&Anemia.
- ↑ with some hormones as: - thyroxine hormone.
- ↑ in exercise → more liberation of O_2 to tissues.
- ↑ in chronic hypoxia as in high altitude & chronic lung diseases to help tissue to get their O_2 needs.
- ↓ in :
  - stored blood
  - acidosis

2- Shift to left:

- At any given PO_2 , Hb affinity to O_2 and % saturation is increased when the curve is shifted to Lt.

i.e, Hb of low affinity to O_2 is changed into Hb with higher affinity → releasing less O_2 to tissues.



Causes of shift to Left:

1- ↓ H+ (Alkalosis)
2- ↓ PCO_2 (hypocapnea)
3- ↓Temperature
4- CO poisoning
5- ↓2,3 DPG
6- Fetal Hb

Fetal Hb:

- It is one of the causes of shift to Lt. → higher affinity of fetal Hb to maternal O_2 which is low in amount.

Hypoxia

Definition: Decreased O_2 delivery to tissues.

Types:

1-Hypoxic Hypoxia: ↓ O_2 delivery to alveoli as in:

- High altitude
- Mines المناجم
- Diseases of respiratory system (Hypoventilation):
  - Resp. center depression. → eg: morphine toxicity.
  - Resp. nerve supply lesion. → eg: Poliomyelitis which attacks the anterior horn cells of intercostal nerve.
  - Resp. muscles lesion → eg: myasthenia gravis.
  - Chest wall deformity → eg: Kyphosis – lordosis
  - Lung diseases → eg: Pneumonia – Pleural effusion & Emphysema
  - Low V/Q ratio.
  - Shunting of venous blood to arterial blood
    - → eg: ventricular- septal defect (VSD)
    - (non-oxygenated blood pass directly to left side of the heart)

NB

- **Hypoxic hypoxia is associated by cyanosis.**
- **It is the only type of hypoxia where hypoxemia develops.**
- **It is the type of hypoxia where treatment by O2 therapy is the most useful among other types of hypoxia.**

2-Anemic Hypoxia:

- **↓ Hb available to carry O_2 as in severe anemia or CO poisoning. (the affinity of Hb to CO is 240 times the affinity to O_2)**

3-Stagnant Hypoxia: (Slow circulation)

Generalized	Localized
As in heart failure (HF) or Polycythemia.	As in intra – arterial thrombosis.

4-Histotoxic Hypoxia:

- **↓ O_2 delivery to tissues due to ↓Cytochrome oxidase enzyme, This occurs in Cyanide Poisoning.**

N.B: Hypoxemia: ↓ O_2 tension (PO_2) in the blood.

Hypoxemia → is present in hypoxic hypoxia only.

	Hypoxic Hypoxia	Anemic Hypoxia	Stagnant Hypoxia	Histotoxic Hypoxia
Arterial O2 content	↓	↓	Normal	Normal
Venous bl. O2 content	↓	↓	↓	↑
Sympt. Signs	Cyanosis	-Anemia (pallor) -CO → Carboxy Hb (cherry red colour)	Cyanosis	-More red than Normal -Hb is full of O2 & can't be delivered to tissues.
Brain Heart GIT Lungs	• Confusion-Headache-coma-mental changes & disorientation. • Tachycardia-hypertension-↑COP(cardiac output). • (ANV) Anorexia- Nausea- Vomiting. • Tachypnea →→ Periodic breathing			
TTT	O2 Therapy	TTT of the cause Anemia→ bl. Transf. CO poisoning→give hyperbaric O_2 & artificial respiration.	TTT of the cause	Nitrate & Meyhylene blue Change Hb to Met Hb in cases of cyanide poisoning MetHb+Cyanide→ Cyanomet Hb(less toxic).

Acclimatization to high altitude

Definition: Gradual improvement of hypoxic effects on the body due to adaptive mechanisms.

N.B: One can tolerate high altitude up to

10,000 feet if he breathes mixed air.

,PO2 is now about 60 mmHg → stimulation of peripheral chemoreceptors

Above these limits→ ↓PO_2 < 40 mmHg

↓ Hb saturation < 75% (Steep part of the curve)

Adaptive Mechanism:

1-Stimulation of Resp. Center

Because hypoxia → stimulate chemoreceptors → tachypnea & Hyperventilation.

N.B:

- this tachypnea ↓ after few days because the alkalosis (developed due to CO_2 washout of tachypnea)
  Lead to Respiratory center inhibition.
- During the period of Alkalosis
  HCO_3^- excretion in urine → Alkaline urine → Correction of alkalosis → Re-stimulation of hyperventilation.

2-↑ O_2 – Carrying capacity of blood

- Hypoxia → + Erythropoiesis because it ↑Erythropoietin hormone . This Leads to
  - ↑ Hb
  - ↑ RBCs > 8 million/
  - ↑ PCV → up to 60%
  - ↑ total blood Vol. (20-30%)

3-↑Circulatory Rate

- Hypoxia → + Chemoreceptors → + Vasomotor Center in MO → (↑COP & ↑Blood pressure) & tachycardia.

4-↑O2 Liberation to tissues

- ↑2,3 DPG (shift of O_2 – Hb dissociation to Rt.)
- ↑Mitochondria
- ↑Myoglobin
- ↑Cytochrome oxidase

N.B:

- At first there is shift of the O_2 – Hb dissociation curve to Lt. due to alkalosis. then there is shift of the curve to Rt. due to ↑2,3 DPG.

Disadvantages of compensatory mechanisms to high altitude:

1- Increased blood viscosity → increased ABP
2- Increased peripheral vascular resistance to blood flow
3- Increased cardiac work to pump blood

- **Acute mountain sickness:** In the 1st few days 4-8 days of the presence in the high altitude. It's effects are:

  -Brain: headache – dizziness – confusion - , ...

  -Heart: tachycardia – palpitation - ↑Bp

  -Lungs: tachypnea

  -GIT: nausea and vomiting

  it is due to cerebral edema & pulmonary edema and alkalosis.

- Acclimatization (adaptation) occurring after these days is called **Chronic mountain sickness.**

Cyanosis

Definition: bluish discoloration of the skin & mucus membrane due to ↑ reduced Hb >5 gm %

N.B

Normally there is a limited amount of reduced Hb
In arterial blood → 0.45 gm %
In venous blood → 3.75 gm %
In capillaries → average = 2.1 gm%

Causes

(1) Hypoxic Hypoxia & Asphyxia

Because in this case the O_2 decreases all over the body &
↓ O_2 content in the blood
→ ↑ reduced Hb in (arteries & veins)

(2) Stagnant hypoxia

↓ O_2 content of venous blood

(3) Polycythemia

- Because of high amount of nonfunctioning Hb.
- It causes stagnant hypoxia.

Types

a- Central: generalized cyanosis specially in the tongue, lips
Occurs in heart failure & cyanotic heart disease & in respiratory failure.

b- Peripheral: internal bed of fingers as during exposure to cold temperature.

Factors affecting cyanosis

1- Skin thickness: it is deeper in thin skin.

2- Skin color (pigmentation):

Dark color (as in Nigro) → masks cyanosis

Yellow races → change the blue color of cyanosis → difficult diagnosis.

3- Capillary factor

Capillary VD as in heat exposure: improve peripheral cyanosis but worsen central cyanosis.

4- Blood composition

e.g. Met-hemoglobinemia gives a color which is cyanosis like.

N.B

- Cyanosis is easier in polycythemia because of ↑amount of non-functioning (reduced) Hb.
  While in Anemia it becomes very difficult because it is difficult to find 5 gm of reduced Hb with low Hb content in anemia
- Very cold weather causes ↓ cyanosis because of vasodilatation occurring due to accumulation of metabolites during ischemia (caused by the very cold weather)

O_2 Therapy

Indications

1- Most beneficial in cases of hypoxic hypoxia .
2- CO poisoning .
3- May be used in cyanide poisoning.
4- May be used to improve cyanosis in stagnant hypoxia .
5- Hyperbaric O_2 may be used in gas gangrene / open heart surgeries .

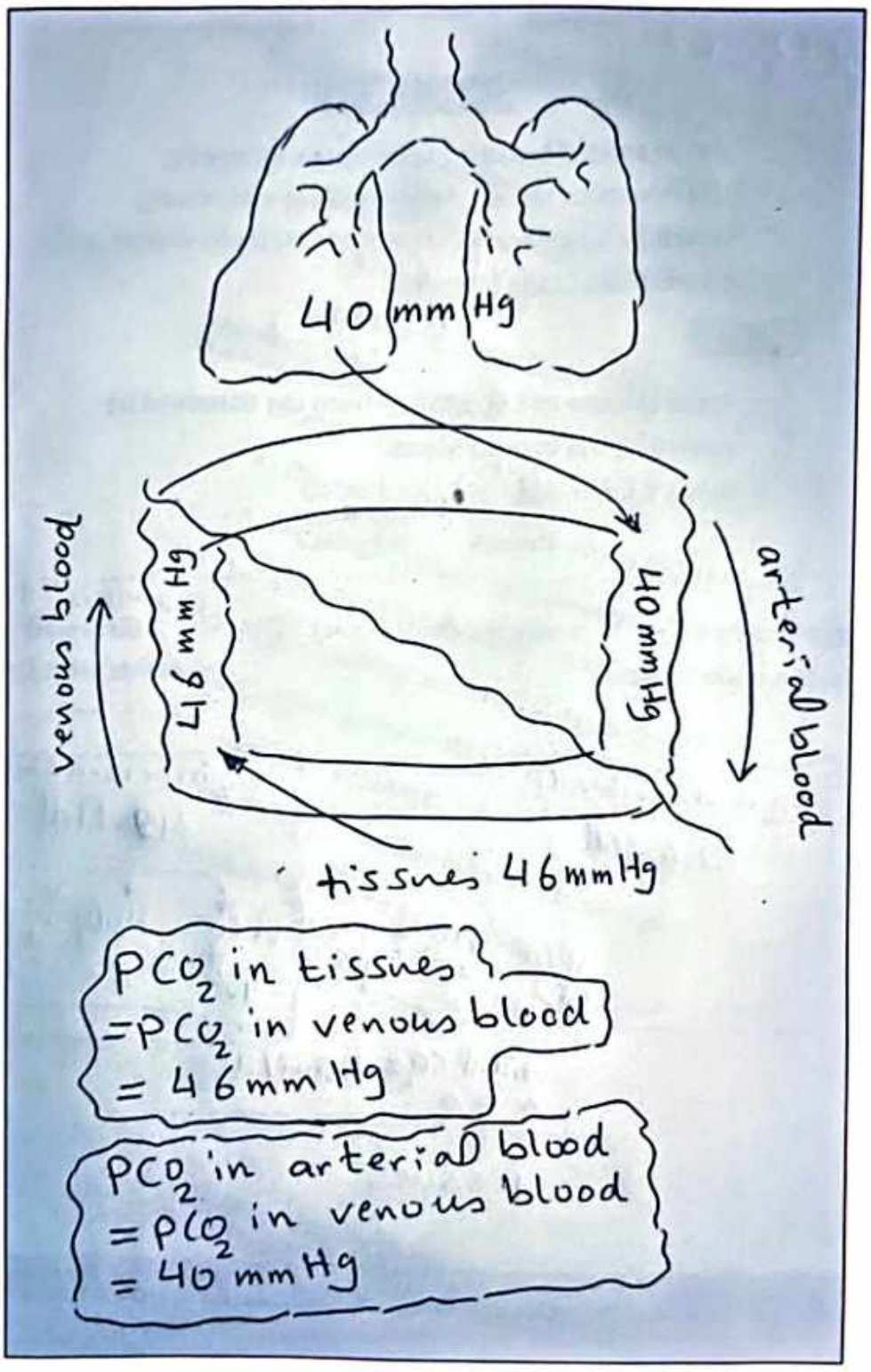
O_2 toxicity and dangers

- Mostly due to O_2 free radicals & H_2O_2.
- start after 8 hours when 100% pure oxygen is used.
- start after few minutes when hyperbaric oxygen is used.

1- Neonatal irritation, sore throat, cough.
2- Trachea-bronchial irritation.
3- Thickening of alveo-capillary membrane → ↓ diffusion
4- ↓surfactant → RDS.
5- 👁 retrolental fibroplasia → blindness .
6- Acidosis → because ↑ O_2 Therapy → ↑ dissolved O_2 →satisfy the tissue → no reduction of oxy Hb →no carriage of CO_2 from the tissue → Acidosis.

CO_2 TRANSPORT

- PCO_2 in arterial blood = PCO_2 in lung = 40 mmHg
- PCO_2 in venous blood = PCO_2 in tissues = 46 mmHg



Tidal Co_2

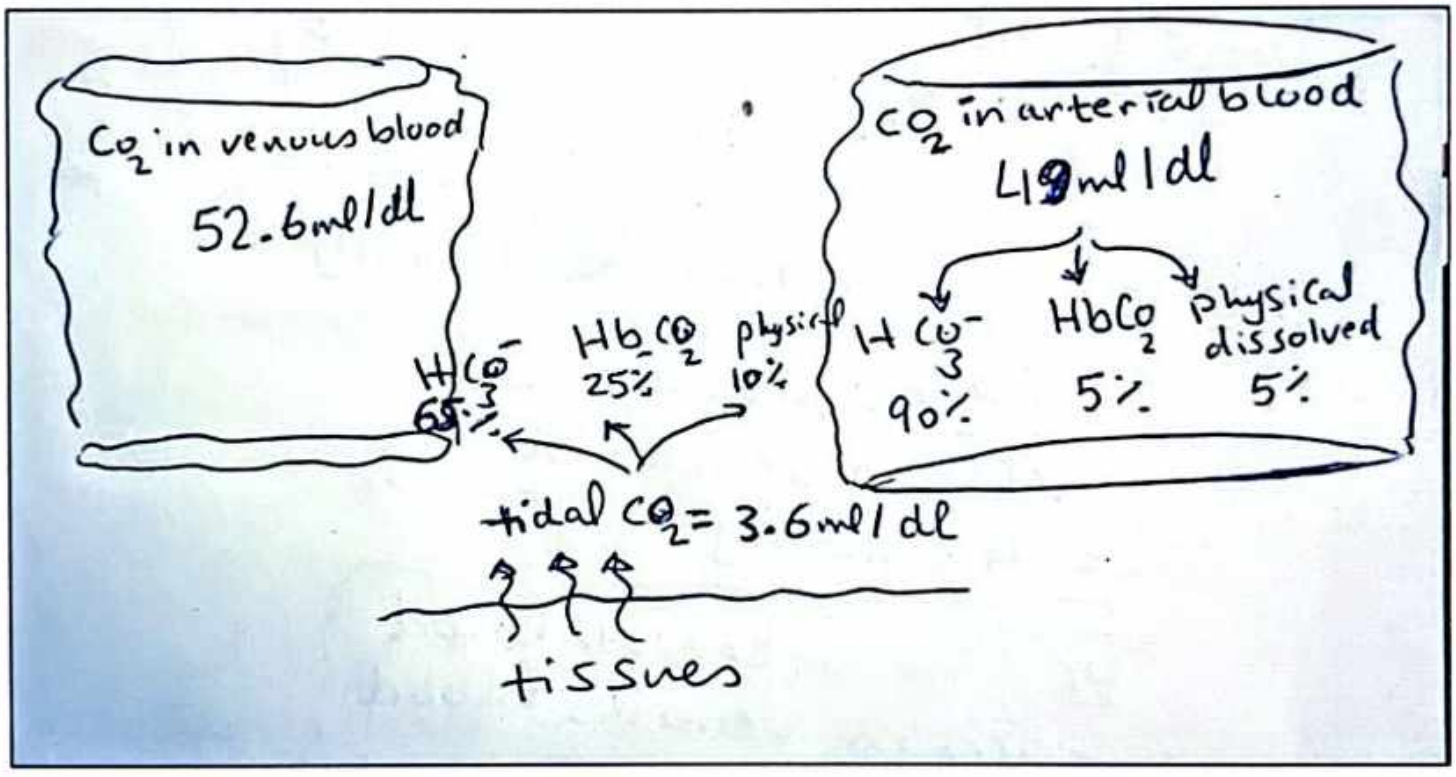
- It the volume of CO_2 diffused from the tissues to be carried by the venous blood.
- Value 3.6 ml / 100 ml of blood

Carried as follow

Bicarbonate 65 % It is the main form

carbamino compound 25%

physically dissolved =free form 10%



Forms of CO_2 in arterial blood

Physical form

- Small volume.
- About 5% of its total content in blood
- Dissolved in H_2O
- Its volume depends on PCO_2 in blood

Chemical form

a- Bicarbonate

- The main form of CO_2 in blood (90%)
- Can be found in 2 components:

1- Inside RBCS bind with K^+ → $KHCO_3$

2- In plasma bind with Na^+ → $NaHCO_3$

b- Carbamino compounds (5%)

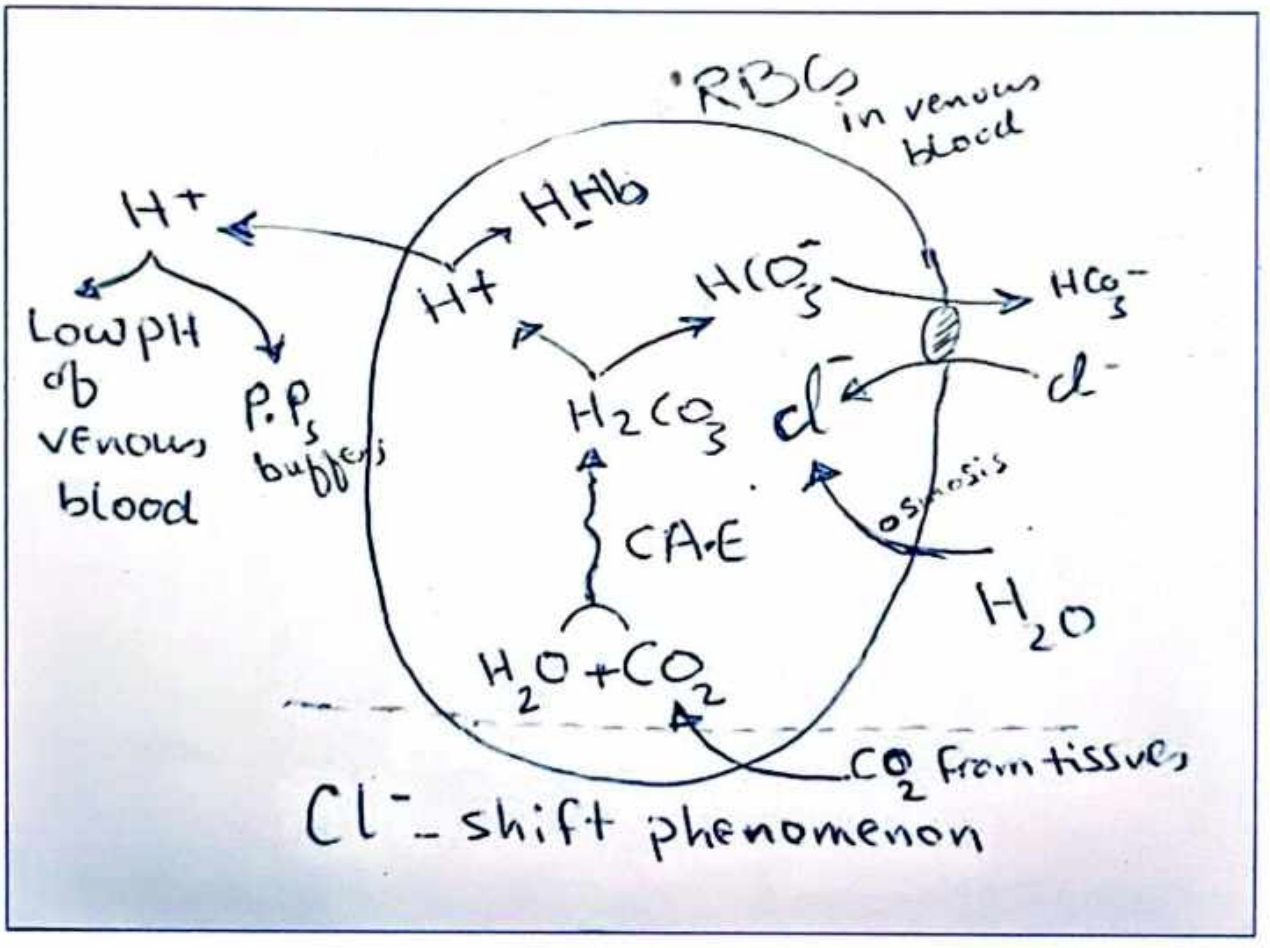
- Bind with proteins:

▯ Inside RBCs → carbamino haemoglobin

▯ In plasma bind to plasma proteins

Chloride shift phenomenon
=Hamburger's phenomenon

- It is the movement of Cl^- in and out of RBCs to compensate for the movement of bicarbonate ions between RBCs and plasma.
  - It results from carrying of most of tidal CO_2 in the form of bicarbonate.
  - Its aim is to maintain electrical neutrality of RBCs
  - It occurs both in the tissues and lung but in reversed manner.



1. At the tissue level

1- CO_2 diffused to inside RBCs and most of it will be changed into carbonic acid H_2CO_3 by the help effect of carbonic anhydrase enzyme (CA).

2- H_2CO_3 will be dissociated into (H^+, HCO_3^-)

3- So, the conc. of HCO_3^- inside RBCs is ↑↑↑ and part of it will pass into plasma according to concentration gradient.

4- Passage of HCO_3^- from RBCs into plasma occurs in exchange with (Cl^-) chloride ions which pass from plasma into RBCs according to the electrical gradient

5- Passage of Cl^- to inside RBCs causes ↑↑↑ of its osmotic pressure, so H_2O will pass from plasma → into RBCs according to osmotic pressure

Results of Cl^- shift phenomenon on the venous blood

	RBCs	Plasma
HCO_3 concentration	↑	↑
Cl^- concentration	↑	↓
H_2O content	↑	↓
PH		More acidic venous blood compared to arterial blood.

N.B

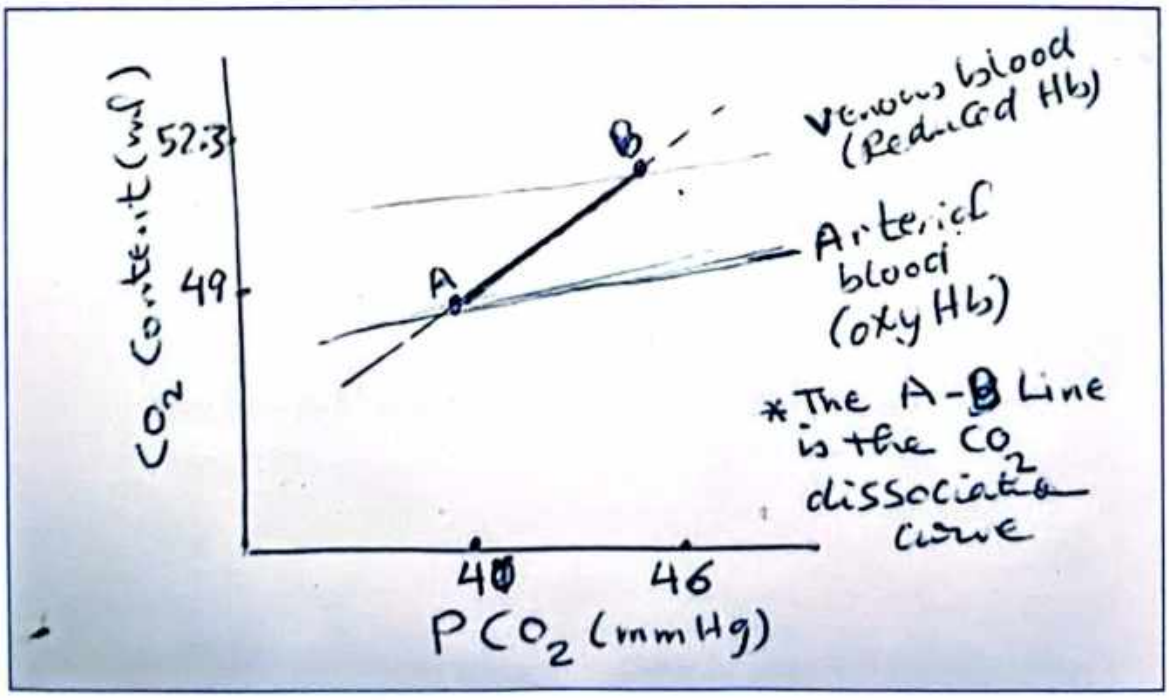
- ✓ Movement of H_2O from plasma → RBCs makes the hematocrit value higher in venous blood compared to arterial blood

2. At the lung

- Cl^- shift phenomenon occurs but **in opposite direction**
- HCO_3^- moves from plasma → RBCs
- Cl^- moves from RBCs → plasma
- H_2O moves from RBCs → plasma

CO_2 dissociation curve

- It is a curve describes the relation between PCO_2 and CO_2 content in the blood.



From the curve we can conclude that:

- At any PCO_2 reduced Hb (venous blood) can carry more CO_2 than oxygenated Hb

  i.e. Deoxygenation of Hb ↑↑↑ its affinity to combine with CO_2.This is called **Halden's effect**
- This helps loading of venous blood with CO_2 released from tissues , and unloading of arterial blood from CO_2 at the lung.
- Physiological CO_2 dissociation curve is represented by the line connecting the points (A) & (B).

Regulation of respiration

control of breathing

(1) Chemical — **(2) non chemical**

More important — less important

(1) Chemical regulation

- It is a chemoreceptor reflex

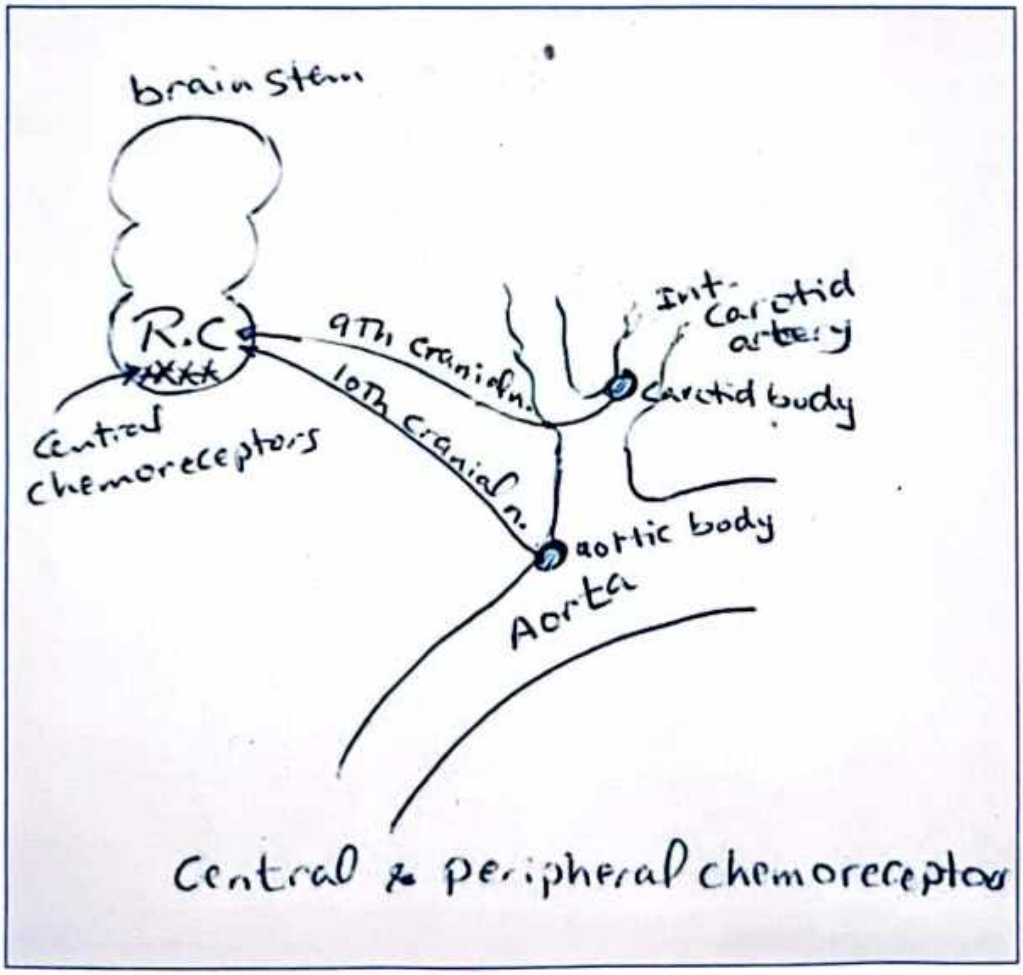
□ Stimulated:-

- Hypoxia (↓ PO_2)
- Hypercapnia (↑ CO_2)
- Acidosis (↑ H^+)
- ↓ ABP

- Nicotine & K^+ also may stimulate the peripheral receptors.

Receptors		
	Central chemoreceptors	Peripheral chemoreceptors
Site	At ventrolateral aspect of MO Inside the blood brain barrier	1. At aorta near the arch 2. At carotid body near the bifurcation of common carotid artery
Sensitivity	↑ CO_2 only Which then form carbonic acid $CO_2 + H_2O \rightarrow H_2CO_3$ which dissociate into HCO_3^- , H^+ H^+ then stimulate the receptors **N.B** H^+ in acidosis can't cross BBB. So,Central chemoreceptors are insensitive to acidosis. *So, the only stimulant to central chemoreceptors is hypercapnia*	Sensitive to all types of stimuli - ↓ PO_2 < 60 % - Acidosis - ↑PCO_2 starting from 35 mmHg and ↑ discharge gradual with increased PCO_2 **N.B** Not sensitive to anemia because in anemia there is ↓ed bound form of O_2 and not dissolved O_2 which forms the PO_2

		** The receptors in *carotid body are more sensitive than* those of aortic arch
Structure	Star shaped ☆☆	2 types of cells * Type 1 (glomus cell) * Type 2 (supporting cell)
Blood supply	Low 1 ml / gm / min	High 20 ml / gm /min
Afferent	No afferent (direct stimulation)	A. body → 10th nerve C. body → 9th nerve



Central & peripheral chemoreceptor

Response

1- Ventilatory response to PCO_2

- PCO_2 is the main driving force for respiration
- PCO_2 >35 mmHg → (+) peripheral chemoreceptors 9th & 10th afferents → (+) R.C – or stimulate the central chemoreceptors in higher PCO_2 levels which cause more stimulation of R.C → hyperventilation→ wash out of the excess CO_2

N.B

- 70 % of the response is due to stimulation of central chemoreceptor.
- 30 % of the response is due to stimulation of the peripheral chemoreceptors.
- In severe hypercapnia → depression of CNS & RC (Headache, Convulsions , Coma, Death)

N.B

- In severe hypercapnea (CO_2 narcosis), treatment of hypoxia is dangerous because hypoxia is now the only stimulant of respiraton (peripheral chemo-receptor effect)

2- Ventilatory response to hypoxia

↓ Po_2 markedly < 60 mmHg. (normally PO_2=100 mmHg → (+) peripheral chemoreceptors only → (+) RC hyperventilation.

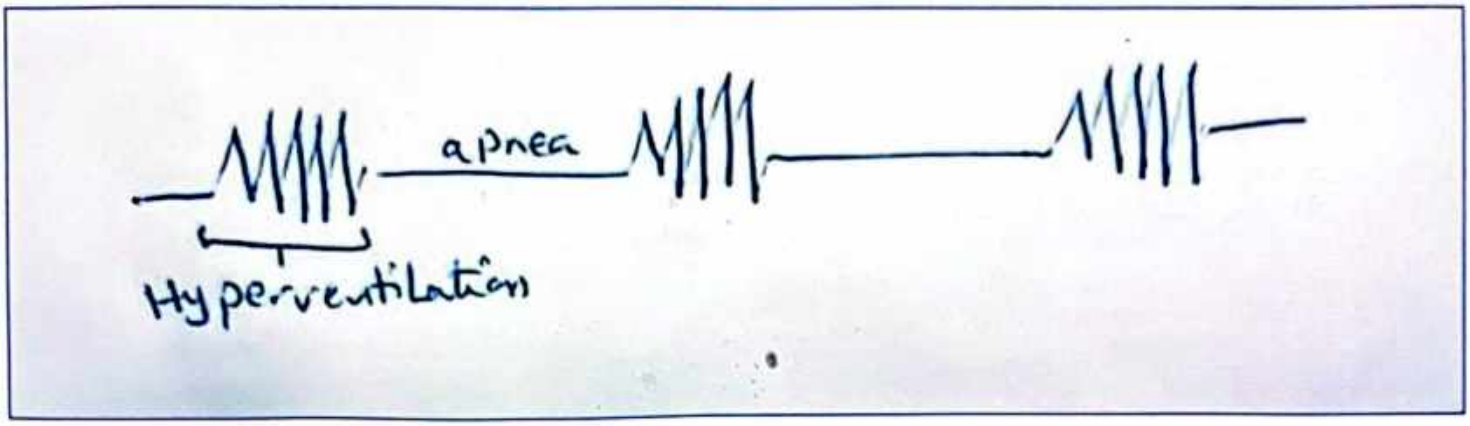
3- Ventilatory response to acidosis

= Kussmaul breathing = air hunger

Acidosis causes very strong stimulation to respiratory center through peripheral chemoreceptors

Cheyne – stoke's respiration

- A case of periodic breathing, (hyperventilation alternating with a period of apnea).



Causes:

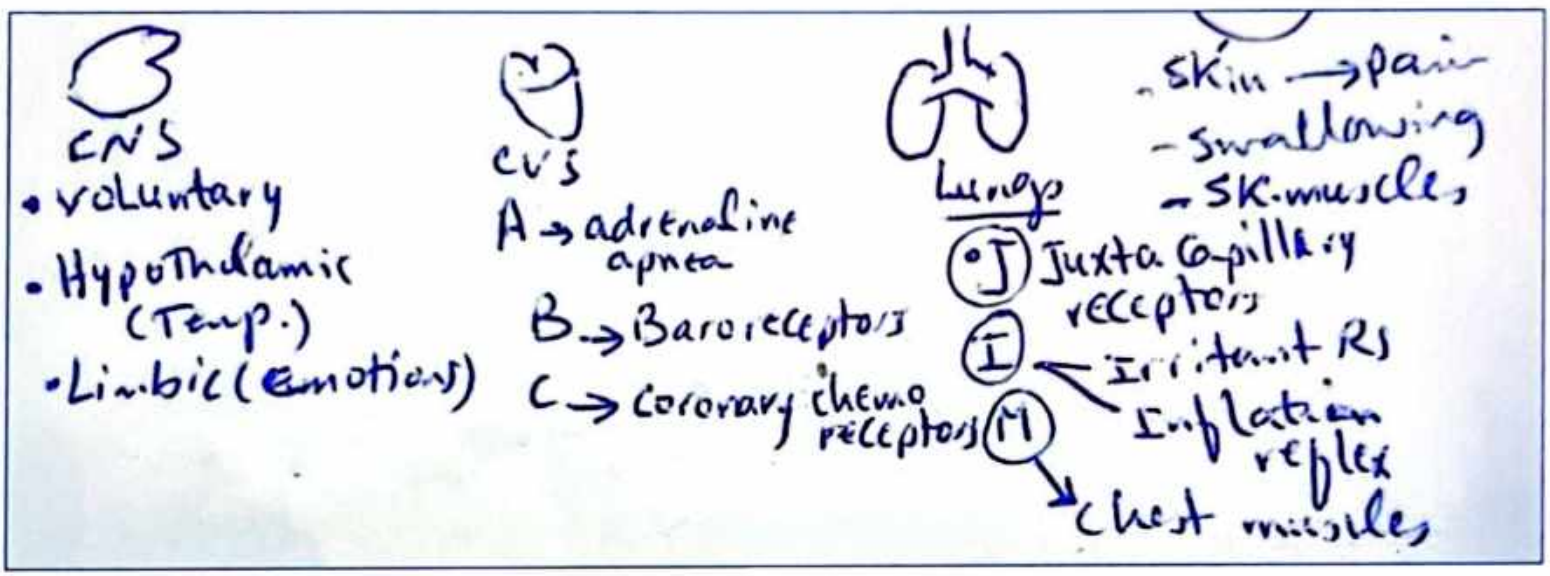
1- increased sensitivity of central chemoreceptors to Co_2 at normal CO_2 → hyperventilation → wash out of Co_2 → apnea till Co_2 level reach its normal value again → hyperventilation→ and so on .....

2- Decreased sensitivity of the chemoreceptors to Co_2 as in:

- Anesthesia
- Morphine poison
- Deep sleep

normal PCO_2 → No stimulation of receptors →apnea →hypoxia →(+) peripheral chemoreceptors→ Hyperventilation → wash out of Co_2 → apnea →and so on .....

2) Non chemical regulation of respiration:



Cortical motor area (CMA)

Voluntary control or inhibition is transient and opposed by the stronger effect of hypercapnea.

- The point at which the voluntary inhibition is stopped is called the **breaking point** Due to accumulation of CO2 and H+ in blood

Hypothalamus → for thermoregulation

e.g. in hot weather → (+) RC → hyperventilation (the expired air causes loss of some of body temperature)

in cold weather → (-) RC → hypoventilation.

- **Limbic system**

  Emotions which are controlled by the limbic system can stimulate or inhibit the RC.

- **J receptor** = **Juxta capillary receptors**

  Close to pulmonary vessels & stimulated in hyperventilation & pulmonary edema or embolism→ rapid shallow breathing.

- **IRRITANT RECEPTORS**
  - Nose → sneezing
  - Larynx → cough & laryngeal spasm
  - Trachea & bronchi → cough & mucus
  - Lung → hyperventilation
- **INFLATION RECEPTORS**

INFLATION RECEPTORS(Hering Breuer reflex)هام

Stim:

✓ Over distension of alveoli during inspiration

Receptors:

✓ stretch receptors in alveolar wall

Afferent: vagus nerve

Response:

✓ inhibition of RC → stop inspiration →start expiration

N.B

- **This reflex is Herring Breuer inflation reflex, while HB deflation reflex occurs upon marked deflation of alveoli during expiration which causes stimulation of RC.**

(M)

Chest muscles & ligaments

Contain receptors which cause Rhythmic Breathing when stimulated.

(A)

Adrenaline apnea

Large dose of adrenaline IV → sudden ↑ in BP → stimulation of baroreceptors → inhibition of RC →apnea

(B)

Baroreceptors stimulated by → ↑ ABP

In aortic arch & carotid sinus → 9th & 10th nerves → (-) RC → ↓ respiratory rate

(C)

Coronary blood

Contain coronary chemoreceptors which are stimulated by nicotine / toxins of myocardial infarction→ causes (-) of RC

Proprioceptors → in muscles, ligaments & joints stimulated in exercise → (+) RC → tachypnea

Skin → mild pain → tachypnea & Severe pain → depression of RC.

Swallowing → receptors in pharynx → (-) RC as during eating or vomiting.

Respiratory center

- It is a collection of neurons located in the brain stem.
- Its function is to regulate respiration via controlling muscles of respiration through descending tracts in the spinal cord.

It is classified into

Medullary center		Pontine center	
DRG contains mainly inspiratory (I)neurons	VRG conatin I & E	Apneustic center	Pneumotaxic center
(its discharge causes normal tidal inspiration	It discharges during forced breathing	Lower pons	Upper pons
DRG are driven by pacemaker cells called pre Bodzinger complex (PBC).present at upper medulla			

Medullary center	
Inspiratory	**Expiratory**
Containing I neurons Its discharge → contraction of diaphragm and other inspiratory muscles i.e. induce respiration	Containing E neurons which are active only during forced Expiration Its discharge → contraction of accessory expiratory muscles → forced expiration
Both centers have reciprocal innervation One stimulus: + of inspiratory center - of expiratory center	

Pontine center

A. Apneustic center

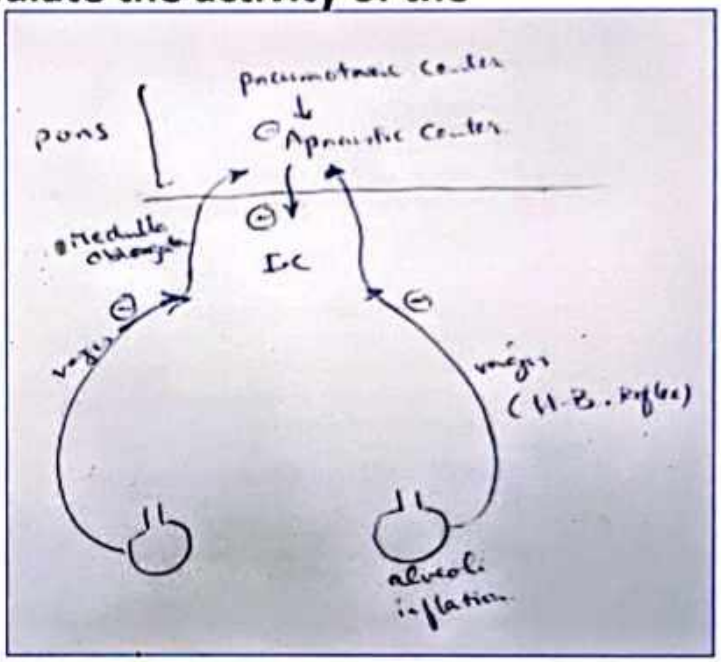
- Found in lower part of pons
- It has inherent tonic activity which causes (+inspiratory center, - expiratory center)
- Its activity is modulated (on & off inhibition) by pneumotaxic center.

NB: Inspiratory center is therefore inhibited by:

1) It is intermittently inhibited by vagal discharge arises from lung which appear during inflation of lung & disappears during deflation of lungs.
2) Pneumotaxic center inhibitory effect on apneustic center.

B) Pneumotaxic center

- Found in the upper part of pons
- Its function is to modulate the activity of the apneustic center on and off inhibition → The normal rhythm of respiration.



N.B

✓ Factors contribute in the generation of the normal breathing rhythm:

1- The rhythm activity of pneumotaxic center
2- The rhythmic intermittent inhibitory discharge from the vagal nerve to the apneustic center

The effect of brain stem transection on respiration

Level of lesion	Effect on breathing	
	Vagus intact	Vagus cut
Upper border Of pons	No effect	Deep slow breathing
Mid pons	Deep slow breathing	Apneustic breathing (only inspiration interrupted by very small expiration)
Lower border of medulla	Stoppage of breathing	

N.B

Apneutic breathing (MCQ)

- It occurs in the form of prolonged inspiratory spasm alternating with weak expiration

Causes

- Transections of brainstem at midpons with bilateral vagotomy.
- Where the inherent tonic activity of the apneustic center is neither modulated by the pneumotaxic center nor inhibited by vagus nerve.

Changes in ventilation during muscle exercise

1- Ventilation is increased suddenly 20 times (from 6L to 120 L), with the onset of exercise (within seconds) then it becomes gradual.

a- The sudden increase is due to:

- Psychic stimulation of respiratory centers and cortical motor areas that stimulate exercising muscles.
- Afferent impulses from proprioceptors in muscles, tendons and joints even before the onset of respiration as a feed forward mechanism (anticipatory mechanisms).

b- The gradual increase is due to:

- Increased body temperature → stimulation of respiratory center
- Increased plasma K + during exercise → stimulates peripheral chemoreceptors.
- Fluctuation of blood gases lead to hypoxia (decreases PO2), and hypercapnia (increased PCO2), they stimulate peripheral chemoreceptors.
- Increased sensitivity of respiratory centers to CO2
- Sympathetic stimulation and increased epinephrine, → increased ventilation
- In vigorous exercise also →anerobic metabolism (increased lactic acid) which stimulates peripheral chemoreceptors.

- In moderate exercise → increased depth of ventilation

- In severe (strenuous) exercise → increased depth and rate of respiration.

2- Arterial PCO_2 → relatively constant or minimal decrease (due to high ventilation rate)

3- Arterial PO_2 → relatively constant or minimal increase (due to high ventilation rate).

4- Arterial H^+ → relatively constant in mild or moderate exercise, but increase in heavy exercise

5- O_2 consumption is increased from 250 ml/ min during rest up to 4000 ml/min

6- CO_2 excretion is increased from 200 ml/min to 8000 ml/min .

7- Increased O_2 delivery to muscles due to
   a- Increased CO_2
   b- Local muscle vasodilation

8- Increased O_2 extraction by exercising muscles due to
   a- Increased PO_2 gradient between tissues and systemic capillaries.
   b- Shift of O_2- Hb dissociation curve to right because of increased CO_2, H^+, 2,3 DPG, and temperature.

After the end of exercise, (recovery period):

- Pulmonary ventilation returns to basal value after the end of exercise by about 90 minutes due to
- Abrupt decreases of ventilation followed by gradual decrease.

- The respiratory rate after exercise doesn't reach basal values until the O2 debt is repaid.
- The high H+ levels (but not due to changes of PO2 or CO2 levels

The extra O2 consumption during the recovery of muscle is used to:

- Resaturate muscle myoglobin by O2
- Resynthesis of muscle ATP and creatine phosphate
- Removal of lactic acid to glycogen (80%) or metabolized into CO2 and H20 (20%)

▪Decompression sickness

Disease characterized by

↑Nitrogen bubbles in blood.

▪This occurs in divers when they rapidly ascend from deep spaces of the sea to the surface.

▪N2 bubbles cause interruption of the circulation which may block the blood flow to important organs.

▪Treatment:

Immediate recompression in pressured chamber followed by slow decompression.